Glance, then Select: Evidence-Tempered Visual Token Selection for Extremely Compressed Multimodal LLMs

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Abstract

Multimodal large language models (MLLMs) encode a high-resolution image into thousands of visual tokens, which dominate the cost of the prefill stage. Visual token reduction removes most of these tokens, but two problems remain at extreme budgets. The vision encoder still processes every patch of the image, and the few surviving tokens often miss the small regions that hold the answer. In this paper, we propose Glance-ETD, a two-stage token reduction framework guided by a lightweight question-conditioned scorer. In the first stage, a glance encoder views the image at low resolution and re-encodes only the question-relevant regions at native resolution. In the second stage, Evidence-Tempered Diversity (ETD) selects the final tokens by combining diversity with the evidence predicted by the scorer, so that the kept tokens cover the scene and retain the answer. A single-rollout self-distillation recipe trains the MLLM with the glance encoder. With only 5% of the visual tokens, Glance-ETD retains 91.6% of the full-token performance of Qwen3.5-4B across fourteen benchmarks and outperforms the strongest prior method, LT-OPD, by 17.3 points and the same training recipe with DivPrune by 6.0. It also speeds up the time to first token on 4K images by 3.1× and generalizes to three other MLLMs and to high-resolution document benchmarks, with the largest gains on text-rich tasks.

1. Introduction

Multimodal LLMs [4, 29, 30, 36] spend most of their prefill on visual tokens. A high-resolution image becomes thousands of tokens, while the question about it is often answered in a few words. Visual token reduction [2, 7, 59] removes most of these tokens before the language model, and LT-OPD [26] shows that a model trained on its own rollouts against a full-token copy of itself can work with only a small fraction of them. Keeping one token in twenty is therefore a realistic operating point. We ask what limits a model at such an extreme budget and how to lift that limit.

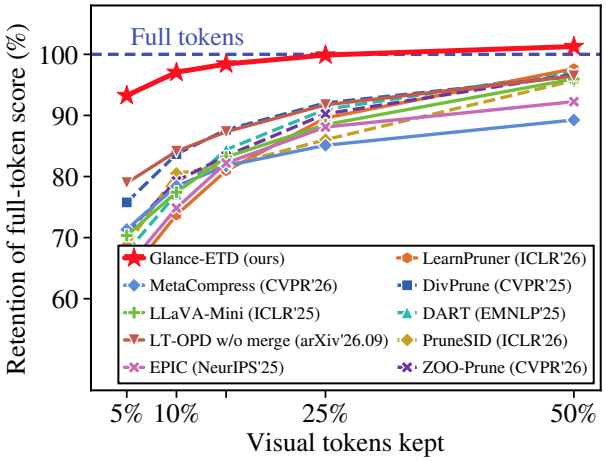


Figure 1. Retention across visual-token budgets on Qwen3.5-4B: the mean over nine benchmarks[1] of each score divided by our non-thinking full-token score, with the same number of tokens per image. Solid lines are trained models; dashed lines are training-free and run on the released model, as in their papers. At 5% of the tokens, Glance-ETD keeps 93.3% (the mean of three training seeds; the other budgets use one seed), 17.5 points above DivPrune.

We find two limits (Fig. 2). The first is cost. Pruning after the vision encoder shortens the language model's prefill, but the encoder still attends over every patch. At high resolution the encoder becomes the dominant stage, so discarding almost all tokens barely changes the time to the first token (Fig. 2a). Reducing tokens only pays off if the encoder also looks at less of the image.

The second limit is selection. When only one token in twenty survives, which tokens survive decides whether the answer is still in the input. Diversity-based selectors such as DivPrune [2] spread the budget over the scene. This works for questions about objects and layout but of-

[1]V* [57], HR-Bench 4K [52], GQA [20], MMMU [61], MMBench (circular) [31], MME [15], POPE [27], TextVQA [46] and OCR-Bench [32].

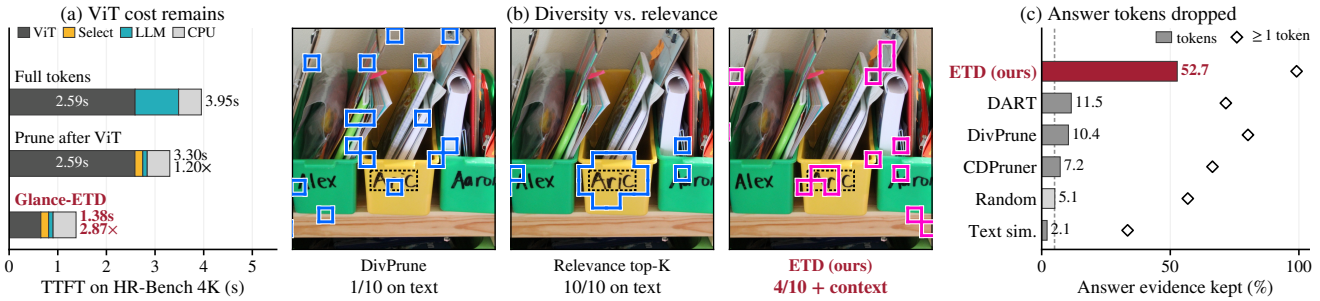


Figure 2. Two problems at a strict 5% visual-token budget on Qwen3.5-4B. (a) Time to first token on HR-Bench 4K for the full-token model, pruning after the ViT, and Glance-ETD (Tab. S14). Each bar is split into vision encoding, selection, LLM prefill and CPU preprocessing. Pruning after the ViT still spends 2.59 s in the encoder; the glance encoder cuts that stage to 0.65 s. (b) One TextVQA image at the same budget: outlines are kept tokens and the dotted box is the answer. DivPrune keeps 1 of 10 text tokens, relevance top-K keeps all 10 but little context, and ETD keeps 4 plus context. (c) On 4,123 TextVQA validation questions with the LT-OPD-recipe model, bars are the share of tokens overlapping the answer OCR box and diamonds the share of questions keeping at least one. The dashed line is the 5% budget.

ten drops the few tokens of a street sign or a phone number (Fig. 2b,c). Selecting by relevance has the opposite failure: it finds the text but loses the context that counting and existence questions need. Training does not remove this limit: a model trained at the budget with the LT-OPD recipe still loses most on reading benchmarks, and changing only the selector recovers much of that loss.

Both limits come down to one question: where in the image is the evidence for this query? If we knew where it lies, we could spend the encoder only on those regions and bias the token selection toward them without losing coverage of the rest of the scene. We build **Glance-ETD** around this idea. A small query-conditioned scorer reads the visual tokens and the question and predicts an evidence map, and the same map guides two stages.

The first stage is a *glance encoder*. It first glances at a low-resolution view and uses the scorer to decide which regions matter. Only those regions are then encoded at native resolution. The rest of the image keeps its coarse encoding, so the output stays on the native token grid and any token selector can follow. The second stage is *Evidence-Tempered Diversity* (ETD). It keeps the greedy diversity loop of DivPrune and tempers each candidate by its evidence. When the evidence map is flat, ETD behaves like DivPrune; when the question points to a small region, the selection moves toward it while still covering the scene. The scorer is trained once on box annotations with the MLLM frozen. A single-rollout self-distillation recipe then trains the MLLM and the glance encoder together, 3.7× faster than LT-OPD training.

The two stages address the two limits directly. On Qwen3.5-4B with 5% of the visual tokens, Glance-ETD retains 91.6% of the full-token score over fourteen benchmarks, 17.3 points above the strongest prior method LT-OPD, 20.3 above DivPrune and 6.0 above our training recipe with DivPrune, and reaches the first token on 4K images 3.1× sooner than the full-token model. It is the best method at every token budget (Fig. 1), with the largest margin at 5%. The gains concentrate on reading and document questions where diversity alone fails. They also carry over to Gemma 4, GLM-4.6V and InternVL3.5 and to further high-resolution and document benchmarks.

In summary, our contributions are as follows.

- We identify two limits of visual token reduction at extreme budgets. Pruning after the vision encoder leaves the encoder cost untouched, and training at the budget does not fix a selector that drops the answer.
- We propose Glance-ETD. Its glance encoder spends native resolution only on the regions that a question-conditioned scorer finds relevant. Its token selector ETD keeps the diversity loop of DivPrune and tempers it by the evidence from the same scorer.
- We run a controlled study at a strict 5% budget over three training regimes, four MLLMs and 14 benchmarks. The study measures the training recipe, the glance encoder and ETD one at a time and shows that the gains come mostly from reading and document questions.

2. Related Work

Training-free token reduction. Most work on visual token reduction leaves the model weights unchanged. One line prunes tokens inside the language model by their text-to-image attention, as in FastV [7], SparseVLM [68] and PyramidDrop [58]. Another line selects or merges tokens before the language model using signals from the encoder, as in VisionZip [59] and PruMerge [39]. DivPrune [2] picks tokens by max–min diversity and ignores the question. CDPruner [64] adds the question to the diversity objective. DART [55] removes duplicated tokens, and VisPruner [65] combines encoder attention with redundancy removal. More recently, ZOO-Prune [21] scores tokens by zeroth-order sensitivity, and PruneSID [14]

removes redundancy within groups of similar tokens. VisionTrim [60] complements attention-based selection with text-guided tokens, and HAWK [71] weights text-to-image attention by head. SPARE [23] measures diversity by subspace reconstruction, and MetaCompress [54] learns a prompt-agnostic compression matrix. All of these methods act after the vision encoder, so the cost of the encoder stays the same. Inside the ViT, token pruning and merging [5, 28, 37] and coarse-to-fine or mixed-resolution encoding [8, 38, 53] were designed for classification; the glance encoder brings the latter to MLLMs with question-chosen regions. FastVLM [49] replaces the encoder with a hybrid one that emits fewer tokens at high resolution.

Training with fewer visual tokens. Training can recover much of the accuracy lost at small budgets. Q-Former [24] and Matryoshka Multimodal Models [6] learn compact or nested sets of visual tokens. LLaVA-Mini [66] compresses the image into a few query tokens before the language model. EPIC [56] distills a compressed student from a full-token teacher step by step. LearnPruner [47] learns a pruning module after the vision encoder and pairs it with text-guided pruning inside the language model. On-policy distillation [1, 18] trains a student on its own samples under a teacher's token-level supervision, extending knowledge distillation [19]. LT-OPD [26] is the closest to our setting. It distills a student on its own rollouts from a frozen full-token copy, with a curriculum over the token budget. We adopt its training recipe and study the token selector and the vision encoder on top of that recipe.

Evidence in high-resolution images. On text-rich images the answer often sits in a few small regions. ETPrune [11] reads question-conditioned evidence from decoder attention, protects text-like regions and sets a budget per sample. It prunes inside the LLM and keeps about half of the tokens. DocPrune [9] removes background and irrelevant blocks from document images. HiRED [3] splits the token budget across high-resolution partitions by encoder attention. ETD works at 5% of the tokens before the LLM. Its evidence comes from a scorer trained on boxes, and its diversity term keeps the scene covered. Visual CoT [40] provides grounded boxes, and we use them only to train this scorer. V* [57] and divide-and-conquer systems [52] search a high-resolution image over several steps. ViCrop [62] crops and FlexAttention [25] selects high-resolution tokens where the MLLM attends, ZoomEye [42] searches a zoom tree, DeepEyes [70] learns to zoom with reinforcement learning, S^2 [43] encodes several scales of the image, and PS3 [44], the encoder behind VILA-HD, is pre-trained to encode a low-resolution view and then only the high-resolution patches the prompt selects. The glance encoder reuses the MLLM's own encoder, adapted by self-distillation, and makes its single pass cheaper on large images.

3. Method

Glance-ETD reduces the visual input in two stages (Fig. 3 and Alg. S1): the glance encoder decides where to look at full resolution (Sec. 3.2), and ETD chooses the K tokens passed to the language model (Sec. 3.3). One small question-conditioned scorer drives both decisions. We then describe the teacher-KL diagnostic (Sec. 3.4) and how the two stages are trained (Sec. 3.5).

3.1. Preliminaries

Problem setup. An image is encoded into N visual tokens $X = \{x_i\}_{i=1}^{N}$ in the LLM embedding space, after the patch merger. Given a question q and a budget $K \ll N$, a selector returns an index set $S \subset \{1, \dots, N\}$ with $|S| = K$, and only X_S is fed to the language model, with its original positions. We use two budget rules. *Strict*: $K = \lceil rN \rceil$ with $r = 5\%$, so that $\sum K / \sum N = 5.0\%$ on every benchmark. *Floor-32*: $K = \max(32, \lceil 0.05N \rceil)$, the rule of the LT-OPD release, which keeps 7.1% of all tokens because small images hit the floor. All selectors compared in this paper keep the same K per image and differ only in which tokens they keep, not in how many visual tokens each image receives.

Diversity selection. DivPrune [2] starts from the token least similar to the mean token and greedily adds

$$i^\star = \arg\max_{i \notin S} d_i, \qquad d_i = \min_{j \in S} \big(1 - \cos(x_i, x_j)\big), \quad (1)$$

until $|S| = K$. The max–min rule, the greedy k-center heuristic [17], spreads the budget over the image and ignores q. ETD keeps it as its diversity term.

On-policy self-distillation. LT-OPD [26] trains a student MLLM that sees only X_S to match a frozen copy of itself that sees all of X. The student generates answers with the pruned input, and the teacher provides token-level supervision along these rollouts, as in on-policy distillation [1, 18]. We adopt this objective for every model trained in this paper as well.

3.2. Stage 1: Glance Encoder

Post-encoder selection leaves the ViT encoder [12] untouched, and for large images the encoder dominates: on HR-Bench 4K, 89% of its FLOPs are global attention over about 50k patches. The glance encoder decides, before full-resolution encoding, which parts of the image the encoder should see in full detail.

Two-pass encoding. (i) *Glance.* The image is average-pooled 2×2 and encoded with full attention at a quarter of the patches, giving one coarse token per 2×2 cell of the native merged grid. (ii) *Look.* The evidence scorer of

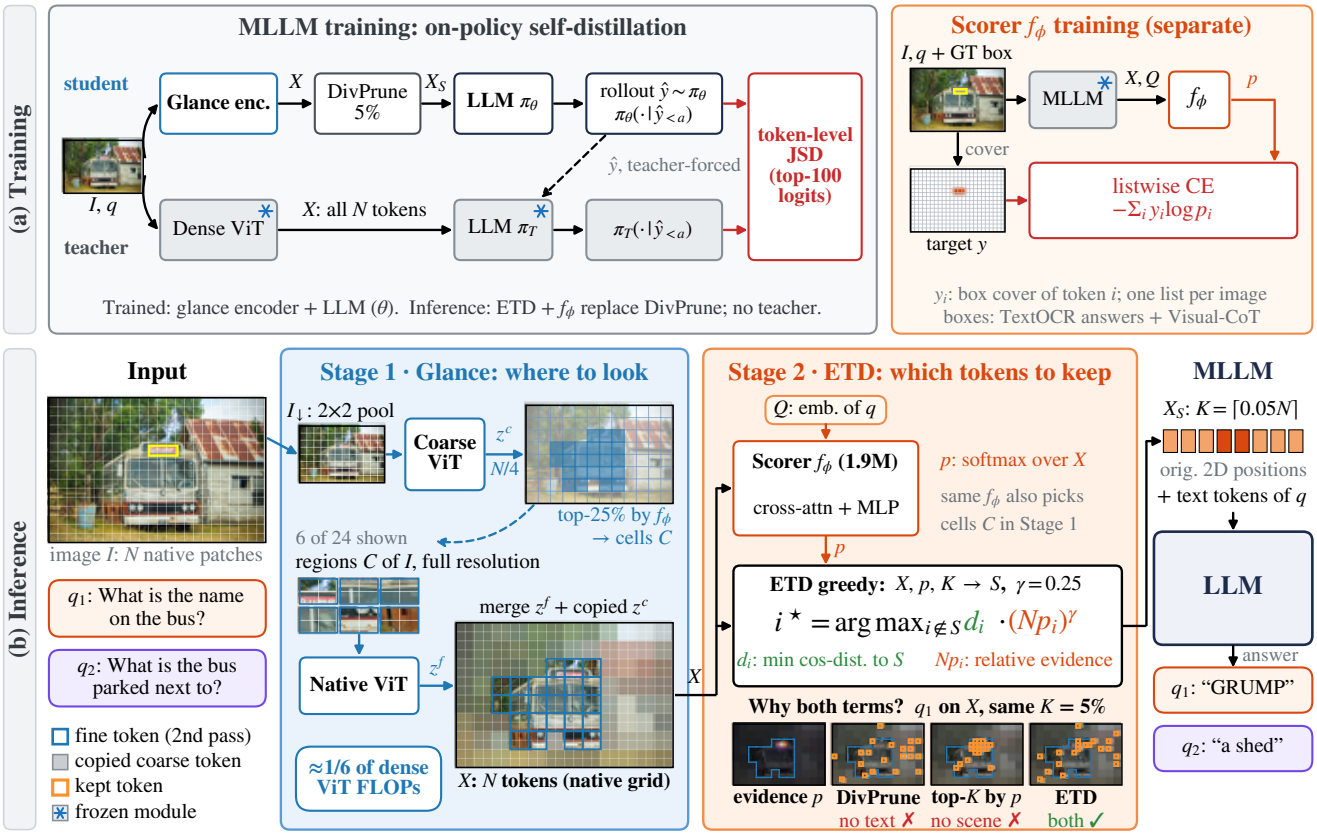


Figure 3. **Glance-ETD.** (a) Training: the student (glance encoder and LLM) selects with DivPrune, rolls out, and is distilled token by token towards a frozen full-token teacher; the scorer f_ϕ is trained separately on box targets with the MLLM frozen. (b) Inference: Stage 1 (Sec. 3.2) encodes the image at a quarter of the patches, the scorer f_ϕ picks the 25% of the coarse cells with the highest question relevance, and only these are re-encoded at native resolution, giving a grid X with all N native positions. Stage 2 (Sec. 3.3) maps X and the question to an evidence map p, and ETD greedily selects K tokens of X by $d_i(Np_i)^{0.25}$ (shown for q_1, on an illustrative evidence map).

Sec. 3.3 reads the coarse tokens and the question, and the 25% of the cells with the highest relevance logits r are selected; the native patches of all selected cells are encoded as one sequence with global attention among them, with the absolute and rotary position codes of their native positions. (iii) *Merge.* After the patch merger, selected cells take their second-pass tokens and every other cell copies its coarse token to its four native slots. The grid has the size of the dense encoder's, so any post-encoder selector applies unchanged, and the LLM gives every slot its own M-RoPE [50] position. The four copies are identical features, so once one is kept the others have $d_i = 0$ and are not selected again. The encoder costs about a sixth of the dense FLOPs, and since the 5% budget is a fifth of the high-resolution pool, the kept tokens come mostly from regions seen at native resolution.

Training. The glance encoder is trained inside the MLLM by the self-distillation objective alone. During training the cells are chosen by max–min diversity (Eq. (1)), so the MLLM never depends on the scorer; the cell choice is discrete and receives no gradient. At inference the scorer chooses the cells. It is applied to images with at least 4096 patches; smaller images are encoded densely at their native resolution.

3.3. Stage 2: Evidence-Tempered Diversity

ETD keeps the diversity loop of Eq. (1) and tilts it towards query evidence. It has two parts: a scorer that produces an evidence map, and a greedy rule that uses it.

Evidence scorer. A small network f_ϕ reads the visual tokens and the question embeddings and outputs a relevance logit r_i for every visual token. Visual tokens are projected to $h = 256$ dimensions and given a 2D sine–cosine position code; one cross-attention layer lets them attend to the projected question tokens, producing a context c_i; an MLP on $[h_i, c_i, h_i \odot \bar{q}]$, with $\bar{q}$ the mean question feature, produces r_i. The scorer has about 1.9M parameters and reads the MLLM's own token features, so no extra encoder is needed; with the glance encoder it reads the coarse grid

in Stage 1 and the merged native grid in Stage 2, the same input that ETD selects from, at about 2 ms per call. The evidence distribution is $p = \mathrm{softmax}(r)$.

Selection rule. ETD replaces the greedy score of Eq. (1) with the evidence-tempered score below,

$$i^\star = \arg\max_{i \notin S} \; d_i \left(N p_i\right)^\gamma, \qquad \gamma = 0.25. \tag{2}$$

The factor Np_i is the evidence of token i relative to a uniform map, and γ is an inverse temperature. If the scorer finds no evidence (p uniform), every weight is one and ETD is exactly DivPrune. A peaked map on a reading question moves the selection to the evidence; a broad map on a scene question leaves it close to diversity. No threshold or question classifier is involved. The value $\gamma = 0.25$ gives the lowest teacher KL on training questions (Sec. 3.4) and is used unchanged for every model and budget. Because the distance term remains, a text region does not absorb the whole budget: once a few tokens of a word are kept, its neighbours have small d_i and the budget moves on. Hard splits, gates and a per-question γ are compared in Sec. 4.4. We compute Eq. (2) in log space with the weights shifted by their maximum, which keeps the ranking exact.

Scorer supervision. The scorer is trained from boxes, not from benchmark labels. For TextVQA training questions we align answer words to TextOCR word boxes; for Visual-CoT [40] we use the annotated evidence boxes, which cover both text and general objects. The target y_i of token i is the fraction of its cell covered by evidence boxes, normalised to $\sum_i y_i = 1$ over the image. The loss is listwise: each image is one list, and the scorer minimises the cross-entropy $-\sum_i y_i \log p_i$ over all its N tokens, so only the ranking of r within an image is supervised. Some images overlap evaluation sets (*e.g.* 105 of 1,284 DocVQA images); a scorer trained without them is in Sec. D.3. The scorer reads the frozen MLLM's tokens of the densely encoded image and its question embeddings, and is trained with AdamW [33] (3×10^{-4}, 16 images per step) for three epochs of 60K images sampled from 422K records (28K TextVQA, the rest Visual-CoT, sampled at 0.3 of the TextVQA weight). It also has a text head used only by the t-based ablations (Sec. D.1); training without it gives the same retention.

3.4. Teacher-KL Diagnostic

To compare selection rules without running benchmarks, we measure how far a subset moves the model's predictions away from the full-token ones. For a full-token greedy answer A of at most 32 tokens, the cost of a subset S is

$$\mathcal{L}(S) = \sum_{a=1}^{|A|} \mathrm{KL}\big(\pi(\cdot \mid X, q, A_{<a}) \,\big\|\, \pi(\cdot \mid X_S, q, A_{<a})\big), \tag{3}$$

computed with teacher forcing along A on the untrained model. On 8,000 training questions at a fixed K, ETD reaches 0.46 of the $\mathcal{L}$ of DivPrune, almost all of the reduction on text questions; among exponents $\gamma \in \{0.25, 0.5, 1, 2, 4, 8\}$, 0.25 gives the lowest $\mathcal{L}$. Fitting $\mathcal{L}$ as a power law in the kept fraction gives a per-image budget $K_m \propto N_m^{0.45}$, analysed in Secs. D.1 and E.3 and not used for our main results.

3.5. Training and Inference

Training recipe. The training recipe is the third part of Glance-ETD. We train Qwen3.5-4B with on-policy self-distillation (Sec. 3.1); the student rolls out with 5% of the tokens selected by DivPrune and keeps LT-OPD's learnable merge of discarded tokens into kept ones. The training loss is the token-level generalised Jensen–Shannon divergence ($\alpha = 0.5$) between teacher and student along the student's rollouts, computed on a top-100 token support with an aggregated tail bucket. We use one rollout per prompt and a 256-token answer cap, which shortens training from 17.4 h to 4.7 h on 8 A800 GPUs at the same accuracy (Sec. B.5). Training uses LT-OPD-14K with 1.5K spatial-reasoning samples replacing the same number of originals, for 175 steps. The vision encoder uses learning rate 2×10^{-6}, ten times the LT-OPD default, so that the glance encoder can adapt to its two-resolution input. The higher rate raises the dense model with DivPrune from 85.3% to 88.8% and Glance-ETD by 3.4 points; Sec. 4.3 reports the recipe and the two stages one at a time, at 10× and at the LT-OPD rate.

Scorer and inference. The scorer is trained once on the features of the untrained MLLM and shared by all Qwen models; retraining it on the features of each trained model does not help (Sec. D.3). At inference, the scorer chooses the glance cells, the glance encoder produces the token grid, the scorer computes p on it, and ETD selects K tokens, which enter the LLM with their original positions. The MLLM is thus trained with diversity selection and evaluated with the scorer. This keeps the MLLM independent of the scorer, so the same rule applies to the untrained model; training with ETD instead is compared in Sec. C.2.

4. Experiments

4.1. Experimental Setup

Models. The main experiments use Qwen3.5-4B [36] in non-thinking mode; Sec. 4.2 repeats the comparison on three other MLLMs. **Glance-ETD** is the self-distilled model with the glance encoder (Sec. 3.5), evaluated with scorer-chosen glance cells and ETD. The *dense-encoder variant* is trained in the same way without the glance encoder and evaluated with ETD, and *untrained* refers to the released model.

Table 1. Comparison with token-reduction methods on Qwen3.5-4B and Gemma 4 E4B-it at a strict 5% budget, where every method keeps the same number K of visual tokens per image. Methods are grouped into training-free methods, learned modules trained with the MLLM frozen (ETD trains only its scorer), and methods that adapt the MLLM. The two self-distillation rows share the same training recipe and differ only in the token selector; Self-distillation + ETD is our method with the dense encoder. Ret. is the score on each benchmark relative to the same model's full-token score, averaged over the fourteen benchmarks. Self-distilled rows report the mean ± std over three training seeds (Tab. S1). ‡GQA answers are lower-cased before scoring.

	General					Fine-grained high resolution					Text and document				Ret. (%)
Method	GQA	MMMU	MMB	MME	POPE	V*	HR-4K	HR-8K	MME-RW	HRScene	TextVQA	OCRB	DocVQA	InfoVQA	All
Training-free								*Qwen3.5-4B*							
Full tokens (100%)	60.0	51.3	79.0	2218	89.9	80.6	78.0	72.6	48.9	62.5	80.5	848	94.6	79.4	100.0
DivPrune [2] (CVPR'25)	45.3	46.3	63.4	1695	77.4	70.2	73.4	70.5	38.1	49.6	49.2	264	32.1	21.7	71.3
DART [55] (EMNLP'25)	45.0	48.2	53.6	1529	77.6	49.7	61.5	60.4	30.1	45.0	49.6	125	24.9	21.8	62.8
CDPruner [64] (NeurIPS'25)	35.7	47.2	59.7	1605	74.3	53.9	62.7	56.4	29.7	43.3	43.3	215	21.8	20.5	61.8
PruneSID [14] (ICLR'26)	44.3	47.0	55.0	1602	77.9	69.6	73.8	68.8	35.0	49.2	25.1	108	22.3	20.7	65.2
VisionTrim [60] (ICLR'26)	41.8	45.8	55.4	1514	69.4	48.7	63.8	59.0	23.9	43.8	20.3	89	15.8	21.5	56.9
HAWK [71] (CVPR'26)	30.7	47.4	49.0	1336	70.9	49.2	60.4	56.2	22.5	44.3	16.5	72	9.7	18.9	52.9
ZOO-Prune [21] (CVPR'26)	46.2	46.4	60.1	1646	81.3	72.3	72.2	67.9	36.8	49.4	26.8	118	25.5	20.8	67.0
Frozen MLLM, learned module								*Qwen3.5-4B*							
LLaVA-Mini [66] (ICLR'25)	43.5	47.9	65.3	1579	75.7	56.5	68.6	61.7	32.4	50.1	34.9	236	34.7	27.4	66.8
LearnPruner [47] (ICLR'26)	43.3	45.6	55.1	1505	72.1	48.2	69.0	60.9	29.2	43.7	28.2	87	12.7	19.3	58.8
MetaCompress [54] (CVPR'26)	41.9	46.9	66.1	1576	69.8	63.4	70.4	62.7	27.8	49.2	41.2	241	28.5	22.9	65.9
ETD (ours)	47.2	46.1	63.2	1772	84.8	78.0	75.8	73.8	42.3	52.6	70.2	426	71.9	36.8	**82.0**
MLLM adaptation								*Qwen3.5-4B*							
EPIC‡ [56] (NeurIPS'25)	42.1	44.4	57.0	1233	66.3	54.5	64.8	59.8	39.8	48.2	50.9	120	25.4	22.9	63.0
LT-OPD [26] (arXiv'26.09) w/o merge	47.4	48.6	66.4	1771	81.9	67.0	69.6	65.1	40.5	51.4	55.8	341	40.6	25.6	74.3
Self-distillation + DivPrune	54.2	48.3	73.5	1988	85.9	75.0	77.1	73.1	45.5	55.3	70.4	489	67.2	35.7	85.6±0.2
Self-distillation + ETD (ours)	55.0	48.8	74.9	2009	88.1	79.1	77.2	73.2	45.6	56.1	76.3	597	83.5	49.6	90.5±0.4
Glance-ETD (ours)	54.7	49.6	75.2	1985	88.2	83.6	78.5	75.9	48.8	56.7	76.3	590	82.6	48.1	**91.6**±0.3
Training-free								*Gemma 4 E4B-it*							
Full tokens (100%)	41.3	46.9	73.6	1686	88.1	58.1	61.1	54.4	42.4	54.6	71.9	764	88.3	62.0	100.0
DivPrune [2] (CVPR'25)	37.1	42.9	60.5	1103	83.9	49.2	53.0	46.1	23.7	43.7	38.9	184	14.7	15.0	66.8
DART [55] (EMNLP'25)	33.2	41.3	52.5	934	72.4	34.6	40.2	35.8	17.9	34.7	19.9	83	9.1	12.3	53.1
CDPruner [64] (NeurIPS'25)	35.3	42.2	54.7	962	78.4	28.8	44.4	38.6	14.3	37.3	28.9	138	10.4	13.5	55.9
PruneSID [14] (ICLR'26)	36.8	41.1	60.2	1164	79.9	27.2	41.2	38.4	21.4	39.7	30.9	246	12.3	16.1	60.1
VisionTrim [60] (ICLR'26)	33.5	38.4	55.7	1046	78.0	41.4	54.4	47.5	22.7	39.6	24.0	138	9.6	15.6	60.7
HAWK [71] (CVPR'26)	30.4	40.2	50.4	874	69.2	21.5	30.3	26.9	10.8	31.3	9.6	85	7.4	13.6	45.1
ZOO-Prune [21] (CVPR'26)	36.5	43.1	60.1	1062	81.8	34.0	41.2	36.6	19.9	39.9	24.9	151	10.6	15.2	58.7
Frozen MLLM, learned module								*Gemma 4 E4B-it*							
LLaVA-Mini [66] (ICLR'25)	30.3	39.9	30.5	1018	65.5	38.7	32.5	35.4	29.2	34.9	11.9	62	8.5	18.3	51.1
LearnPruner [47] (ICLR'26)	32.6	40.9	47.9	952	72.7	28.3	36.3	37.3	14.6	35.3	13.3	25	4.1	12.0	49.4
MetaCompress [54] (CVPR'26)	35.6	40.6	59.4	1063	77.7	27.2	33.9	31.2	17.6	36.8	22.9	196	10.8	16.5	55.0
ETD (ours)	37.7	42.2	63.3	1186	83.3	55.0	62.0	56.0	31.8	44.0	59.9	396	39.1	21.0	**78.6**
MLLM adaptation								*Gemma 4 E4B-it*							
EPIC [56] (NeurIPS'25)	40.5	43.1	55.3	1362	78.2	44.5	47.6	42.2	34.5	41.5	24.5	171	13.7	17.9	66.1
LT-OPD [26] (arXiv'26.09) w/o merge	39.3	43.1	63.5	1176	82.5	37.7	52.2	47.8	27.5	42.6	39.1	349	16.2	16.1	68.7
Self-distillation + DivPrune	41.3	42.9	70.8	1386	87.3	49.7	57.3	53.0	29.3	44.7	56.8	440	26.4	18.6	78.1±0.8
Self-distillation + ETD (ours)	41.2	42.7	71.2	1459	86.2	50.4	60.9	55.5	35.3	46.8	66.1	546	42.7	23.6	**84.2**±0.7
Glance-ETD (ours)	40.8	41.4	70.0	1472	86.1	52.9	61.6	51.0	35.2	47.9	64.7	542	40.0	25.0	83.6±1.4

Benchmarks and metric. Tab. 1 covers fourteen benchmarks: the nine of LT-OPD, V* [57], HR-Bench 4K [52], GQA [20], MMMU [61], MMBench (circular) [31], MME [15], POPE [27], TextVQA [46] and OCR-Bench [32], and HR-Bench 8K [52], MME-RealWorld [69], HRScene [67], DocVQA [34] and InfoVQA [35]. We summarise them by the *retention*, the mean ratio of each score to the same model's full-token score (first row of each block; repeated full-token runs in Sec. B.2), over all fourteen (Ret. All). Ablations and budget sweeps report the retention over the nine LT-OPD benchmarks. Overlap between evaluation images and the scorer's training images is analysed separately in Sec. D.3.

Budgets and baselines. The main budget is strict 5% (Sec. 3.1); floor-32 is in Sec. E.1. Baselines are peer-reviewed methods from 2025–2026, re-run on Qwen3.5-4B with the same K per image as ETD and ported to select after the vision merger (Sec. B.3). LLaVA-Mini, LearnPruner and MetaCompress train their modules on LT-OPD-14K with the MLLM frozen. LT-OPD w/o merge is its recipe retrained on each model without the token merge, which our trained models keep (Sec. B.4).

Implementation. We evaluate with VLMEvalKit [13] and lmms-eval [63] using greedy decoding. MMBench groups that rule matching cannot resolve are adjudicated by fixed, judge-free matching rules (Sec. B.2).

Table 3. Ablation of the design choices in Glance-ETD on Qwen3.5-4B and Gemma 4 E4B-it at a strict 5% budget. Each row changes one choice of the full pipeline: Stage 1 rows change which glance cells are re-encoded, and Stage 2 rows change how tokens are selected. Ret. is the retention over the nine LT-OPD benchmarks (six of which are shown), and Δ is its change from the full pipeline. All rows use training seed 0; three-seed means of seven Qwen rows keep the same order (Sec. D.2).

	Qwen3.5-4B, Glance-ETD								Gemma 4 E4B-it, Glance-ETD							
Setting	V*	HR-4K	GQA	POPE	TVQA	OCRB	Ret.	Δ	V*	HR-4K	GQA	POPE	TVQA	OCRB	Ret.	Δ
Full pipeline	82.7	78.9	53.8	88.2	76.8	583	92.9	–	51.8	63.3	43.6	86.2	65.6	550	92.9	–
Stage 1: which cells to re-encode																
max–min diversity	77.5	76.2	53.9	87.9	76.4	580	91.9	-1.0	47.1	58.0	43.7	86.0	62.1	554	90.8	-2.1
random cells	73.8	74.1	53.9	88.1	76.8	585	90.9	-2.0	47.6	55.9	43.6	86.5	61.8	580	90.9	-2.0
Stage 2: which tokens (same K per image)																
w/o evidence: DivPrune	77.5	79.6	53.3	86.7	70.8	502	90.0	-2.9	51.8	58.6	43.5	87.2	60.4	467	89.7	-3.2
strong evidence bias: $\gamma = 64$	77.5	78.6	51.7	86.7	73.6	523	89.3	-3.6	49.2	58.1	43.8	86.8	61.1	515	90.4	-2.5
$\gamma = 1$: $d_i\, Np_i$	84.8	78.4	52.6	87.8	75.8	578	92.0	-0.9	52.4	59.5	42.8	84.9	63.6	536	90.9	-2.0
$\gamma = 0.125$	78.0	79.2	54.5	87.7	76.4	584	92.4	-0.5	50.3	61.7	44.0	86.5	66.4	536	91.9	-1.0
γ set per question by the text confidence t	80.6	80.2	54.2	87.8	76.2	579	92.2	-0.7	51.3	59.8	43.0	85.9	63.8	540	91.5	-1.4
hard split: top-$K/2$ by p + DivPrune	81.7	78.6	53.4	86.5	75.0	541	91.3	-1.6	50.8	60.4	43.5	86.8	62.6	514	90.3	-2.6
hard gate: split only if $t > 0.5$	77.5	79.2	53.3	86.5	74.7	539	90.8	-2.1	52.4	60.9	43.1	86.7	62.2	512	91.1	-1.8
scorer given a wrong question	80.6	78.8	52.5	87.7	72.7	523	90.7	-2.2	51.8	59.2	43.2	85.6	64.2	504	90.8	-2.1
DPP-r instead of ETD	82.7	78.9	49.3	87.7	76.5	592	91.6	-1.3	49.2	60.9	43.2	85.6	66.2	554	91.5	-1.4

Table 2. Component ablation on Qwen3.5-4B at a strict 5% budget. Glance: glance encoder, otherwise dense. ETD: evidence-tempered selection, otherwise DivPrune. The 10× and 1× LR groups self-distil the MLLM with that ViT learning rate (1× is the LT-OPD recipe); No train is the released model. Ret.: retention over the nine LT-OPD benchmarks, mean ± std over three seeds for trained models. TFLOPs: ViT + LLM prefill on HR-Bench 4K. Shaded: Glance-ETD.

	10× LR				1× LR				No train	
Glance	–	–	✓	✓	–	–	✓	✓	–	–
ETD	–	✓	–	✓	–	✓	–	✓	–	✓
V*	75.0	79.1	74.9	83.6	73.8	76.4	73.1	82.0	70.2	78.0
HR-4K	77.1	77.2	76.8	78.5	77.2	77.2	77.1	78.3	73.4	75.8
GQA	54.2	55.0	54.1	54.7	52.3	53.3	52.0	52.5	45.3	47.2
MMMU	48.3	48.8	50.4	49.6	48.9	48.9	48.3	48.3	46.3	46.1
MMB	73.5	74.9	73.8	75.2	70.6	71.8	70.9	72.1	63.4	63.2
MME	1988	2009	2001	1985	1865	1843	1856	1840	1695	1772
POPE	85.9	88.1	86.2	88.2	85.2	87.8	85.2	87.4	77.4	84.8
TextVQA	70.4	76.3	70.5	76.3	64.7	74.2	64.2	74.1	49.2	70.2
OCRB	489	597	489	590	394	512	396	523	264	426
Ret.	88.8 ±0.2	92.5 ±0.4	89.4 ±0.3	93.3 ±0.4	85.3 ±0.2	89.1 ±0.6	85.0 ±0.3	89.9 ±0.2	75.8	83.8
TFLOPs	373	373	65	65	373	373	65	65	373	373

4.2. Main Results

Frozen MLLM. With the MLLM untouched, ETD keeps 82.0% at strict 5%, 10.7 points above DivPrune, the strongest training-free baseline (Tab. 1); the learned modules of LLaVA-Mini, LearnPruner and MetaCompress stay below DivPrune. Importance-based methods (52.9–67.0%) lose most of the text (TextVQA 17–27). Downscaling each image to the budget retains 72.7%, and adding a native-resolution crop of the scorer's cells gains only 0.3 (Sec. B.3). ETD leads at every budget from 1% to 25%, by the widest margin at 5% (Sec. E.2).

MLLM adaptation. Trained rows are means over three training seeds (Tab. S1). LT-OPD w/o merge retains 74.3% (released, with merge: 80.5%); ETD alone, inserted into the released checkpoint without retraining, raises it to 87.8% (Sec. B.4). Our training recipe retains 85.6% with DivPrune and 90.5% with ETD; the full Glance-ETD system retains 91.6±0.3%, 6.0 points above the recipe with DivPrune. EPIC (63.0%) stays below DivPrune, mostly on the reading benchmarks.

Other MLLMs. On Gemma 4 E4B-it [16] with the MLLM frozen, ETD keeps 78.6%, 11.8 points above DivPrune and above the other training-free methods (45.1–60.7%); with MLLM adaptation Glance-ETD retains 83.6% and the dense-encoder variant 84.2%, against 68.7% for LT-OPD without its merge (Tab. 1).

4.3. Ablation Study

Tab. 2 removes each stage at strict 5%. Within each recipe the four columns differ only in the two components. The 10× encoder learning rate alone adds 3.5 points to DivPrune (85.3→88.8%). At this recipe ETD adds 3.7 (92.5%), mostly on TextVQA and OCRBench (70.4→76.3, 489→597), and the glance encoder adds 0.8 (93.3%) while cutting HR-Bench 4K prefill from 373 to 65 TFLOPs and TTFT from 3318 to 1283 ms. With the LT-OPD recipe ETD adds 3.8 on the dense and 4.9 on the glance encoder, the glance encoder stays within one point of the dense one (85.3 *vs.* 85.0 with DivPrune, 89.1 *vs.* 89.9 with ETD), and the 10× rate adds 3.4 to Glance-ETD. Restricting the dense model's candidates to the glance cells leaves it at 92.2% (Sec. C.2): the glance gain comes from native re-encoding.

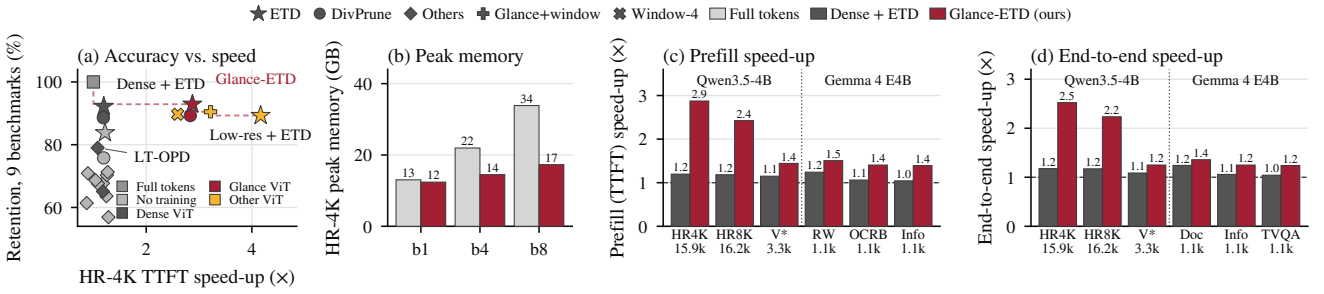


Figure 4. Efficiency at strict 5% (one A800, bf16). (a) Retention (9 benchmarks) against HR-Bench 4K TTFT speed-up; dashed line the Pareto front. (b) HR-Bench 4K peak memory at batch 1, 4 and 8. (c) TTFT and (d) end-to-end speed-up at batch 1 on the three benchmarks with the largest speed-up per model (all in Tab. S15; more in Tab. S14 and Fig. S2).

4.4. Analysis of the Selection Rule

Tab. 3 changes one inference choice of Glance-ETD at a time on the seed-0 models (92.9% on both). We give the drop in retention on Qwen, with Gemma in parentheses; Sec. D.2 describes every row and repeats seven rows over three seeds; the ranking holds on every seed.

The glance should follow the question. Choosing these cells by diversity, the rule used in training, costs 1.0 (2.1), and random cells cost 2.0 (2.0). The loss is on V* and HR-Bench 4K, where the answer is a small object (V* 82.7→77.5 with diversity cells).

ETD needs both evidence and diversity. At $\gamma = 0$ ETD is DivPrune and loses 2.9 (3.2), mostly on reading (TextVQA 76.8→70.8, OCRBench 583→502). At $\gamma = 64$ it keeps the tokens with the highest p and loses 3.6 (2.5): they cover the same words several times and miss the rest of the scene (GQA 53.8→51.7).

How the two are combined. Splitting the budget in half between evidence and diversity costs 1.6 (2.6), switching by question type costs 2.1 (1.8), and the same evidence in the DPP [22] of CDPruner [64] costs 1.3 (1.4), ties ETD only with a tuned temperature, and is four times slower.

A mild tilt is enough. Every γ from 0.125 to 1 stays within one point of 0.25 on Qwen and two on Gemma, so one γ serves every benchmark (Sec. D.2).

The evidence comes from box supervision. With the MLLM frozen, a scorer trained without the question input keeps 5.5 of ETD's 8.0 points over DivPrune, one trained without boxes keeps none (Tab. S7). Giving the scorer another question of the same benchmark costs 2.2 (2.1). Without any TextVQA, TextCaps, DocVQA, InfoVQA or GQA training data the scorer still gives 93.0% (Sec. D.3). On TextVQA, ETD keeps a token inside the answer box for 99.1% of the questions, against 80.2% for DivPrune (Fig. 2c, Sec. E.4).

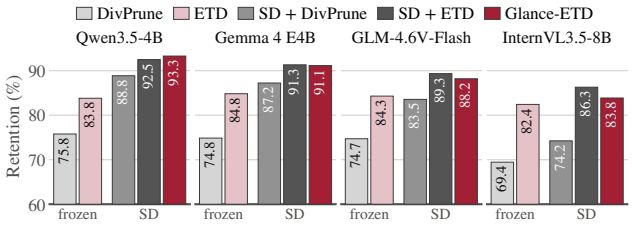


Figure 5. Retention (nine LT-OPD benchmarks, strict 5%) on four MLLMs. SD: self-distillation with the dense encoder (Tab. S13).

4.5. Efficiency

Fig. 4 reports speed and memory (Sec. F). On HR-Bench 4K the ViT is 2.60 s of a 3.96 s TTFT, so selectors after the dense encoder stay at 0.87–1.28× (Tab. S16). Glance re-encodes only scorer-chosen cells, cutting the ViT to 0.57 s and prefill from 482 to 65 TFLOPs: TTFT is 3.09× and end-to-end time 2.69× shorter, batch-8 throughput is 2.94× higher and batch-8 memory halves from 34.7 to 17.7 GB (Tab. S14). The gain grows with image size (Tab. S15): TTFT is 2.42–3.09× shorter on HR-Bench and 1.22–1.47× on V*, DocVQA, HRScene and MME-RealWorld, and 0.87–0.90× on the six benchmarks below the glance threshold. On Gemma, with about 1.1k tokens per image, Glance-ETD shortens TTFT 1.13–1.51×.

4.6. Cross-Backbone Generalization

Fig. 5 repeats the comparison on GLM-4.6V-Flash [48] and InternVL3.5-8B [51], which splits images into crops; each model gets its own scorer and our recipe (Sec. E.6). ETD leads DivPrune on every model, by 8.0–13.0 points when frozen and by 3.7–12.0 after self-distillation. Glance-ETD stays above self-distillation with DivPrune on all four, within 2.4 points of the dense encoder.

5. Conclusion

In Glance-ETD one small evidence scorer decides what the glance encoder re-encodes at native resolution and which tokens ETD keeps. At strict 5% it retains 91.6% on Qwen3.5-4B, 17.3 points above LT-OPD, reaches the first token on 4K images 3.1$\times$ sooner, and leads DivPrune on all four MLLMs, most on reading and documents.

LLM Usage Statement

We used Claude Code (Anthropic, Claude models) as a coding and writing assistant. It wrote and ran parts of the training, evaluation and analysis scripts under the authors' direction, generated the LaTeX tables from the evaluation outputs with scripts, and drafted and revised the manuscript text, which the authors edited and checked. It did not generate experimental measurements: every reported number comes from evaluation runs of the models, and MMBench adjudication is rule-based without a judge model. Citations were checked against the source papers by the authors. No generative model is an author of this paper.

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Glance, then Select: Evidence-Tempered Visual Token Selection for Extremely Compressed Multimodal LLMs

Supplementary Material

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A. Glance-ETD inference

Alg. S1 is the inference procedure. Stage 1 runs on images with at least 4096 patches; smaller images are encoded densely. The same scorer f_ϕ is called twice, on the coarse grid and on the merged grid. The first token is the one least similar to the mean token, as in DivPrune, so a uniform evidence map leaves the loop identical to DivPrune. On a glanced image X has the size of the dense encoder's output: unselected cells contribute their coarse token at each of their four native slots, and these tokens stay candidates for Stage 2 (budget accounting in Sec. B.2). Each later token maximises Eq. (2) with $\gamma = 0.25$. Training does not use the scorer: the MLLM is distilled with diversity cells and DivPrune, and the scorer is trained separately (Sec. 3.5 describes that training in the main text).

Algorithm S1 Glance-ETD inference.

Require: image, question q, rate r=0.05, γ=0.25
Ensure: K visual tokens at their native positions
if at least 4096 patches **then**
  $\bar{X} \leftarrow$ encode the 2×2-pooled image
  $(\bar{r}_i, \cdot) \leftarrow f_\phi(\bar{X}, q)$
  $C \leftarrow$ the 25% of the coarse cells with largest $\bar{r}_i$
  re-encode the native patches of C
  $X \leftarrow$ second-pass on C, coarse copies elsewhere
else
  $X \leftarrow$ dense encoder output
end if
$(r_i, \cdot) \leftarrow f_\phi(X, q)$
$p \leftarrow \mathrm{softmax}(r)$
$N \leftarrow |X|, \quad K \leftarrow \lceil rN \rceil$
$S \leftarrow \{i$ least similar to the mean of $X\}$
while $|S| < K$ **do**
  $d_i \leftarrow \min_{j \in S}(1 - \cos(x_i, x_j)), i \notin S$
  $i^\star \leftarrow \arg\max_{i \notin S}\ d_i (N p_i)^\gamma$
  $S \leftarrow S \cup \{i^\star\}$
end while
return X_S

B. Protocol and Baselines

The entries below fix the seeds, the scoring protocol, the baseline ports, and the cost of the shared training recipe.

B.1. Training Seeds and Variance

Tab. 1 reports, for our method, the mean over three training seeds. Tab. S1 lists all three. Seeds differ only in the training random seed of each run.

B.2. Evaluation Protocol and Scores

All models are evaluated without thinking: the chat template pre-fills an empty think block and the answer tag. The LT-OPD paper normalises by thinking-mode full-token scores from the model card (e.g. MMMU 77.6), which no non-thinking row can approach; we normalise by one fixed non-thinking full-token run of our own instead. MMMU uses the official 128-token answer cap.

Full-token repeats. Two further full-token runs under the same protocol give V* 80.6 / 80.6 / 80.6, HR-Bench 4K

Table S1. Three training seeds at the strict 5% budget of Tab. 1. The main table reports the mean over the three seeds; the best seed (Ret. All) is in bold.

Method	Seed	GQA	MMMU	MMB	MME	POPE	V*	HR-4K	HR-8K	MME-RW	HRScene	TextVQA	OCRB	DocVQA	InfoVQA	Ret. All
Self-distillation + DivPrune	0	54.0	47.9	73.9	1975	86.3	74.9	77.0	73.5	45.9	56.2	70.5	481	68.4	36.0	**85.7**
	1	54.1	48.3	73.2	2033	85.3	74.9	76.6	72.8	45.2	54.9	70.0	500	67.2	35.6	85.5
	2	54.4	48.7	73.5	1957	86.1	75.4	77.8	73.0	45.3	54.8	70.6	486	66.0	35.6	85.4
Glance-ETD	0	53.8	49.6	75.5	1955	88.2	82.7	78.9	75.0	48.0	57.0	76.8	583	82.8	48.7	91.2
	1	55.0	50.1	75.2	1998	88.1	84.3	78.9	76.0	50.0	55.0	75.5	603	82.4	47.5	**91.8**
	2	55.3	49.1	74.9	2001	88.4	83.8	77.9	76.6	48.3	58.1	76.8	584	82.5	48.0	91.7
Dense-encoder variant	0	55.1	47.9	74.7	1991	88.0	79.6	78.1	74.5	45.3	55.5	75.9	585	83.1	48.5	90.2
	1	55.1	49.7	75.1	2057	88.3	79.1	76.2	72.2	45.3	57.5	76.6	603	83.7	50.4	**90.9**
	2	54.8	48.9	74.7	1979	87.8	78.5	77.2	73.0	46.2	55.2	76.5	602	83.8	49.9	90.4

78.0 / 78.0 / 78.0, GQA 60.0 / 60.0 / 60.0, MMMU 51.3 / 51.3 / 51.2, MMBench 79.0 / 79.2 / 79.2, MME 2218 / 2228 / 2226, POPE 89.87 / 89.93 / 89.89, TextVQA 80.49 / 80.56 / 80.56 and OCRBench 848 / 850 / 847 (the first run is the denominator). Across runs the scores differ by at most 10 on MME, 3 on OCRBench and 0.2 elsewhere. Normalising Glance-ETD (seed 0) by any one run gives 92.88, 92.77 or 92.84% retention, and 92.83% with the mean of the three, so the choice of denominator moves retention by at most 0.1 points.

Each model's own full tokens. Training barely changes full-token accuracy: with every visual token, the LT-OPD-recipe, 10× dense-encoder and glance models (seed 0) reach 101.0, 99.6 and 100.7% of the untrained full-token scores (MMBench $+3$ to $+3.5$, MME -41 to -105). Normalised by each model's own full-token scores, DivPrune / ETD retain 84.4 / 88.6% on the LT-OPD-recipe model (85.4 / 89.6% with the fixed denominator), 89.0 / 92.5% (88.6 / 92.2%) on the dense-encoder model, and 88.6 / 92.2% (89.3 / 92.9%) on the glance model. No entry moves by more than 1.0 point, and the gain of ETD over DivPrune is unchanged.

Retention by benchmark group. Tab. S2 splits the retention of Tab. 1 by its three benchmark groups. Glance-ETD retains 94.1% on the general benchmarks and 99.8% on the high-resolution ones, where V* and HR-Bench are above the full-token model, and 78.1% on text and documents, where the largest absolute gap remains (OCRBench 590 of 848, InfoVQA 48.1 of 79.4, DocVQA 82.6 of 94.6). Under the same training, ETD adds 1.5 and 1.4 points to DivPrune on the first two groups and 13.7 on text and documents, and the glance encoder adds 3.6 on the high-resolution group.

MMBench adjudication. Circular-evaluation groups that rule matching cannot resolve are decided by two fixed rules without a judge model: a group is correct if every answer names the gold letter unambiguously, and wrong if some unparsable answer contains no reading that names the gold letter. Groups matching neither rule would need a judge. No row of Tab. 1 needs a judge. The rules decide at most 8 of the 1,164 groups in any row (0.7% of the MMBench score), and Glance-ETD has at most one such group per seed.

Table S2. Retention (%) by benchmark group of Tab. 1 at strict 5% on Qwen3.5-4B. General: GQA, MMMU, MMBench, MME, POPE. High-res.: V*, HR-Bench 4K / 8K, MME-RealWorld, HRScene. Text: TextVQA, OCRBench, DocVQA, InfoVQA. Trained rows are three-seed means.

Method	General	High-res.	Text	All
DivPrune	81.7	87.1	38.4	71.3
ETD (ours)	84.6	93.2	64.9	82.0
LT-OPD	85.7	85.4	46.2	74.3
Self-distillation + DivPrune	92.5	94.8	65.3	85.6
Self-distillation + ETD (ours)	94.0	96.2	79.0	90.5
Glance-ETD (ours)	94.1	99.8	78.1	91.6

Budget accounting. Tab. 1 and the original LT-OPD code do not use the same budget. The code sets $K = \max(32, \lceil 0.05N \rceil)$, a floor of 32 visual tokens on every image, so the keep rate is not strictly 5% and is greater than 5%. On the nine benchmarks that floor raises the total from 5.0% to 7.1%. The main table removes the floor, $K = \lceil 0.05N \rceil$, and every strict row has $\sum K / \sum N = 5.0\%$. Tab. S6 repeats the selection rules under both. Every selector writes a per-image log of (N, K). On the glance model N counts every slot of the merged grid, including the four copies of each coarse cell, so N and K equal those of the dense encoder on the same image and every method keeps the same number of tokens. Stage 2 may keep coarse tokens: the selector receives all N slots and no quota divides K between second-pass and coarse tokens. The four copies of a coarse cell are identical, so once one is kept the others have $d_i = 0$ and at most one is kept. With 25% of the cells re-encoded, a glanced image therefore offers about $0.25N$ distinct second-pass tokens and $0.75N/4 \approx 0.19N$ distinct coarse tokens to the $K = 0.05N$ budget.

B.3. Baseline Ports from Code

All training-free baselines are ported from the official code and select tokens after the vision merger and before the

LLM. ZOO-Prune perturbs the inputs of Qwen's patch merger, its vision-to-language projector. VisionTrim ranks tokens by the attention they receive in the penultimate ViT block (Qwen's encoder has no CLS token) and fills the rest of its budget with the tokens closest to the question. HAWK scores tokens with the pre-RoPE queries and keys of the first full-attention LLM layer, since the first layers of Qwen3.5 use linear attention; its head weights are recomputed for this layer, following the procedure given in Sec. C.2.

The three learned modules are trained on LT-OPD-14K with the MLLM frozen and select before the LLM, like every other row. LearnPruner trains its 0.53M pruning module with the next-token loss and its ratio penalty (AdamW, 10^{-3}, batch 32, two epochs); only its first stage is used, since its second stage prunes inside the LLM and would pass more than K tokens to the first layers. EPIC finetunes the whole model for one epoch (LLM 2×10^{-5}, encoder 2×10^{-6}, global batch 128) with its token-consistency distillation and a compression ratio annealed to 5%; its DART compressor is moved in front of the LLM so that the strict budget counts the same tokens.

Downscaling without pruning. The simplest way to meet a token budget is to shrink the image until it yields the budgeted number of tokens and keep all of them, which also lowers the encoder cost. Resizing every image for the untrained model to 5% of its visual tokens ($\sum K / \sum N = 4.99\%$) retains 74.9% on the nine LT-OPD benchmarks and 72.7% on all fourteen, against 75.8 / 71.3% for DivPrune and 83.8 / 82.0% for ETD. Downscaling keeps the page layout and is strong on documents (DocVQA 67.8, DivPrune 32.1, ETD 71.9), but it loses small objects and small text (V* 54.5 *vs*. 78.0, TextVQA 42.0 *vs*. 70.2 for ETD). At 25% it retains 93.9% on the nine benchmarks, between DivPrune (92.0%) and ETD (95.7%).

Cropping the evidence. A crop-and-zoom baseline gives the untrained model, within the same K, a native-resolution crop around the cells that Stage 1 of Glance-ETD picks (the max-sum window of $a \times b$ merged tokens over the rank-weighted cells) and the whole image downscaled to the remaining tokens; images that do not take the glance path are only downscaled. It retains 75.2% on the nine benchmarks (downscaling 74.9%, ETD 83.8%), gaining only on V* (57.6 *vs*. 54.5) and HR-Bench 4K (68.5 *vs*. 67.0), and 61.0% on the five additional benchmarks against 68.7% for downscaling and 78.9% for ETD. The smaller global view costs most on documents and scenes (DocVQA 52.6 *vs*. 67.8, HRScene 42.8 *vs*. 50.1): finding the evidence is not enough when the rest of the image becomes a thumbnail.

B.4. LT-OPD Baseline Reproduction

The LT-OPD w/o merge rows of Tab. 1, the LT-OPD curve of Fig. 1 and the LT-OPD row of Tab. S9 are the LT-OPD-14K recipe with the learnable token merge turned off, trained on each model. CDPruner still selects the tokens. In the released checkpoint, each discarded token is assigned to its nearest retained token by cosine similarity, and an MLP adds a bounded residual, initialized at zero, onto that retained token. We disable this merge, so the language model receives the selected tokens alone (pure index pruning). Our self-distilled models keep the merge, as the released checkpoint does (Sec. 3.5); every row passes the same number of tokens to the language model, but the training-free rows and this one carry no merged residual. The released Qwen3.5-4B checkpoint keeps the merge. At strict 5% it retains 85.4% on the nine LT-OPD benchmarks (V* 73.3, HR-Bench 4K 71.5, TextVQA 63.3, OCRBench 419), and at floor-32 91.1% (V* 73.3, HR-Bench 4K 71.6, TextVQA 63.6, OCRBench 572); our reproduction with the merge retains 90.3% at floor-32. On the five added benchmarks at strict 5% it scores HR-Bench 8K 67.8, MME-RealWorld 44.4, HRScene 54.7, DocVQA 48.9 and InfoVQA 27.4; on documents it is below the LT-OPD recipe with DivPrune (DocVQA 52.5) and well below Glance-ETD on that benchmark (DocVQA score 82.0).

ETD on the LT-OPD checkpoints. Both LT-OPD checkpoints select with CDPruner. Replacing only the selector at inference, with no retraining, the released checkpoint retains 85.4% (CDPruner), 86.2% (DivPrune) and 89.8% (ETD) on the nine benchmarks, and 71.6 / 76.2 / 84.2% on HR-Bench 8K, MME-RealWorld, HRScene, DocVQA and InfoVQA (TextVQA 63.3→75.3, OCRBench 419→506, DocVQA 48.9→76.2, InfoVQA 27.4→42.0). Our reproduction without the merge (LT-OPD w/o merge in Tab. 1) gives 79.0 / 81.1 / 87.7% on the nine benchmarks and 65.9 / 71.5 / 83.6%. ETD therefore improves LT-OPD as trained, without our recipe or the glance encoder.

Released checkpoint on all fourteen benchmarks. Tab. S3 lists both LT-OPD variants. Over the fourteen benchmarks the released checkpoint retains 80.5% at strict 5% with its own CDPruner selection and token merge, 82.6% with DivPrune and 87.8% with ETD, against 74.3% for the row without the merge. Glance-ETD (91.6%) is therefore 17.3 points above LT-OPD without its merge, 11.1 above the released LT-OPD model, and 3.8 above the released model with ETD inserted at inference. Across the budgets of Fig. 1 (5/10/15/25/50%, nine benchmarks), the released checkpoint retains 85.4/89.7/92.6/94.8/98.3% and LT-OPD without its merge 79.0/84.1/87.3/91.7/96.5%.

From LT-OPD to Glance-ETD. On the nine benchmarks the path from the LT-OPD row to Glance-ETD is 79.0% (LT-OPD, CDPruner in training and inference) → 81.1% (same weights, DivPrune at inference) → 85.3% (trained with DivPrune, one rollout, 256-token cap and the 1.5K spatial samples; Tab. 2, 1× LR) → 88.8% (10× encoder learning rate) → 92.5% (ETD) → 93.3% (glance encoder). Of

Table S3. LT-OPD with and without its learnable token merge on Qwen3.5-4B at a strict 5% budget (CDPruner selection, as trained). The released checkpoint keeps the merge; w/o merge is the LT-OPD-14K recipe retrained with the merge turned off, the LT-OPD row of Tab. 1 and Fig. 1. Ret. as in Tab. 1.

	General					Fine-grained high resolution					Text and document				Ret. (%)
Method	GQA	MMMU	MMB	MME	POPE	V*	HR-4K	HR-8K	MME-RW	HRScene	TextVQA	OCRB	DocVQA	InfoVQA	All
LT-OPD variants								*Qwen3.5-4B*							
Full tokens (100%)	60.0	51.3	79.0	2218	89.9	80.6	78.0	72.6	48.9	62.5	80.5	848	94.6	79.4	100.0
LT-OPD [26], released (with merge)	52.8	49.8	73.5	1895	85.2	73.3	71.5	67.8	44.4	54.7	63.3	419	48.9	27.4	80.5
LT-OPD w/o merge	47.4	48.6	66.4	1771	81.9	67.0	69.6	65.1	40.5	51.4	55.8	341	40.6	25.6	74.3
Glance-ETD (ours)	54.7	49.6	75.2	1985	88.2	83.6	78.5	75.9	48.8	56.7	76.3	590	82.6	48.1	**91.6**±0.3

Table S4. Training cost of the self-distillation recipe (LT-OPD-14K, 175 steps, 8 A800 GPUs; CDPruner selection at the floor-32 budget of the LT-OPD release). Retention here uses a full-token reference with MMMU 52.4 instead of 51.3, which lowers every value by about 0.2 points relative to the main tables; the comparison within the table is unaffected.

Rollouts per prompt, answer cap	Training time	Ret. (%)
8, 1024 (LT-OPD default)	17.4 h	90.1
2, 1024	8.7 h	91.1
1, 1024	7.3 h	90.5
1, 256 (ours)	4.7 h	90.7
1, 128	4.3 h	90.3

the 14.3 points, 2.1 come from DivPrune at inference, 4.2 from the step to our LT-OPD-recipe models, 3.5 from the encoder learning rate, 3.7 from ETD and 0.8 from the glance encoder. The 4.2 points are not the recipe changes. With the token merge kept on and DivPrune at inference (seed 0), our LT-OPD reproduction trained with CDPruner retains 85.0%, and replacing CDPruner by DivPrune in training (86.0), adding the spatial samples (85.1), using one rollout (86.2) and the 256-token cap (85.5) keep it within 1.2 points. The gap to 81.1% separates these merge-trained models from the one retrained without the merge.

B.5. Self-Distillation Training Cost

Tab. S4 varies the number of rollouts per prompt and the answer cap. One rollout with a 256-token cap trains in 4.7 h instead of 17.4 h without a loss in retention, and we use it for all trained models. The amount of data is not a bottleneck either: training on coverage-selected subsets of LT-OPD-14K with the same recipe keeps 90.0–90.7% from 700 samples (5%) up to 7K samples, against 90.4% with all 14K. The 1.5K spatial samples that replace LT-OPD-14K rows are multiple-choice counting, depth and left–right or above–below questions, 750 on synthetic indoor scenes and 750 generated from COCO instance annotations; COCO images used by GQA, POPE or MME are excluded. The LT-OPD row of Tab. 1 is trained on the original LT-OPD-14K.

C. Other Models and Encoders

The same selectors are measured with a reinforcement-learned scorer and with the other vision encoders reported in this section.

C.1. Reinforcement Learning of the Scorer

We trained the scorer with GRPO [41], using the negative teacher KL of the selected subset as a label-free reward and the split rule as the policy, together with a budget head. At strict 5% this gives 88.6% (88.7–88.8% with the budget head), against 87.7% for the supervised scorer under the same rule and 89.6% for ETD. Variants with group Plackett–Luce sampling, token-level credit and expert iteration reached 86.1–87.4%, and GRPO with the greedy subset as the baseline of the advantage reached 88.5%. The best of 16 sampled subsets, chosen by teacher KL, reaches 96.6%: the reward is informative, but online RL closes about one point of the 8.9-point gap. Distilling the best-of-16 subsets offline into the scorer gives 87.7%, and 87.3% when the scorer also reads query/key features of the frozen LLM, so offline distillation does not close the gap either.

C.2. Vision-encoder variants

Tab. S5 compares encoders trained with the same recipe and the same 10× encoder learning rate, all with ETD and, for the glance encoders, scorer-chosen cells. Relative to glance with the same auxiliary loss at strict 5%, encoding everything at low resolution loses 13.7 points on V* and 6.3 on HR-Bench, and window attention loses 10.3 on HR-Bench, where it cuts the global context of 4K images. A second-pass window (glance25w) keeps V* but loses 8.9 on HR-Bench (90.5% *vs*. 91.9% over nine benchmarks). Over three seeds the glance encoder of the paper is 0.8 points above the dense encoder at strict 5% (93.3±0.4 *vs*. 92.5±0.4), ahead on V* and HR-Bench 4K and level on MMBench and MME. To separate the candidate set from the encoding, we restrict Stage 2 on the dense model to the glance pattern: every token of the top-25% scorer cells and one token per other 2×2 cell, on images of at least 1,024 merged tokens. In a separate three-seed evaluation this changes the dense model's retention from 92.3 to 92.2% (five held-out

benchmarks: 61.7 to 61.8), while the glance model reaches 93.0% (62.4). The gain therefore comes from re-encoding the chosen cells at native resolution, mostly on V* (+4), HR-Bench 8K (+3) and MME-RealWorld (+2.5).

The encoder variants are trained with the same recipe as the glance encoder and differ only in the encoder path for images of at least 4096 patches. *Low resolution* encodes the 2×2-pooled image only and copies every coarse token to its four native positions (glance with no second pass). *Window-4* keeps the dense input but restricts attention to non-overlapping windows of 8×8 merged tokens in all blocks except four global ones (5, 11, 17, 23), as in Qwen2.5-VL [4]. *Glance25w* is the glance encoder whose second pass uses the same windowed attention. These variants use the auxiliary loss described next. On MMMU the three-seed means are 48.9 (glance), 48.8 (dense, 10× LR) and 48.9 (dense, 1× LR), so the encoder choice does not change the MMMU result here.

Auxiliary loss. An earlier version of the glance encoder added a cosine loss (weight 0.1) between its merger output and that of the dense encoder. Without it, Glance-ETD retains 93.3±0.4% over three seeds instead of 92.0±0.1%, and DivPrune on the glance model 89.4% instead of 87.7%. The difference is on MMBench (75.2 *vs*. 72.7) and MME (1985 *vs*. 1933), whose images are below the glance threshold and take the dense path: the loss held the shared encoder close to the frozen dense features. The paper's glance encoder is trained without it. The encoder variants of Tab. S5, the zoom-ratio, threshold and feature analyses of Sec. D.4 and the deduplicated-scorer runs of Sec. D.3 use the earlier encoder; each compares settings on the same weights.

Training with ETD. We also self-distil the glance model with ETD (constant exponent 0.25) instead of DivPrune selecting the student's tokens, over three seeds. With Glance-ETD at inference it retains 93.5±0.9%, against 93.3±0.4% for the model trained with DivPrune; with DivPrune in Stage 2 (Stage 1 unchanged) it retains 89.0±0.2% against 90.3±0.4%, lower on reading (TextVQA 67.6 *vs*. 71.0, OCRBench 464 *vs*. 505). ETD adds 3.0–4.5 points at inference whichever selector trained the model, so we train with DivPrune and keep the MLLM independent of the scorer.

HAWK head weights on Qwen3.5. The released HAWK head weights belong to Qwen2.5-VL-7B. We recompute them for the first full-attention layer of Qwen3.5-4B with HAWK's procedure (Eq. 3 of [71]): each of the 16 heads in turn is blocked from all visual tokens on the full-token model, and the drop in gold-answer log-likelihood is measured on 300 questions from each of five LT-OPD-14K training subsets. The weights range from 0.042 to 0.123; head 13 is the most important in three of the five subsets. At strict 5% the recomputed weights and uniform weights give the same retention (56.7% both), so the HAWK row of Tab. 1 does not depend on this choice.

D. Selection Behavior

These measurements separate which tokens are kept from how many, on the training images and on the evaluation benchmarks.

D.1. Teacher-KL Selection Analysis

Which tokens. At a fixed K on 8,000 training questions, the total cost $\mathcal{L}$ of Eq. (3) under ETD is 0.46 of DivPrune's: 0.40 on text questions and 1.03 on general ones, and ETD lowers the KL on 58% of the questions. Over constant exponents $\gamma \in \{0.25, 0.5, 1, 2, 4, 8\}$ the ratio rises monotonically from 0.46 to 0.64, so 0.25 is the best of these exponents on training questions; on 2,000 validation questions of the scorer data with the glance model the KL is flat between 0.25 and 0.5 (1.035 and 1.031). The hard split (top-$K/2$ by p, the rest by diversity) reaches 0.64. Choosing the exponent per question by an oracle gives 0.28, and the oracle exponent grows with t (Spearman $\rho = 0.24$; mean 0.68 and 0.82 in the two lower quartiles of t, 1.39 in the two upper ones), but following that trend with $\gamma = t$ gives 0.50, above the constant. Learned temperature heads on t, the concentration of p, $\log N$ and other features did not improve over a constant either on the dense model at the floor-32 budget (Tab. S6).

How many tokens. On training questions the per-image KL follows a power law in the kept fraction, $\mathcal{L}_m(K) \approx c_m (K/N_m)^{-\alpha}$ with $\alpha = 0.89$. Minimising $\sum_m \mathcal{L}_m(K_m)$ under $\sum_m K_m = B$ equalises the marginal gains (water-filling); with c_m independent of N_m this gives

$$K_m \propto N_m^{\eta}, \qquad \eta = \alpha/(\alpha+1) \approx 0.47. \tag{S1}$$

Fitting η directly on training questions gives 0.45, and this single power law matches an 8-bin lookup table fitted on the same data (validation KL 0.85 *vs*. 0.83). Large images receive a smaller fraction because their KL is already low at 5%. The confidence t does not help predict the budget; answer length does, and a factor of two for open-ended prompts, also fitted on training prompts, completes the rule used in Tab. S11 on this model.

Text head. Besides r_i, the scorer's MLP outputs a text logit s_i, trained jointly by a binary cross-entropy (weight 0.2) on whether the cell touches an OCR word, on records with OCR boxes. It defines the *text confidence* $t = \sum_j p_j \, \sigma(s_j) \in [0, 1]$, the share of the evidence map on text. Glance-ETD uses neither s nor t; they serve only the gated and t-scaled variants of Sec. 4.4, and a scorer trained without the text loss gives the same retention (Sec. D.3).

Calibration of the text confidence. We check the scorer's text head on data it was not trained on: TextVQA

Table S5. Vision-encoder variants, all trained with the same LT-OPD recipe and 10× ViT learning rate except the first row. Low-res, window and glance variants except the last row add the auxiliary cosine loss (Sec. C.2). Seed 0, selection at inference only.

Method	V*	HR-4K	GQA	MMMU	MMB	MME	POPE	TextVQA	OCRB	Ret. (%)
DivPrune, strict 5%										
Dense ViT	73.8	76.2	52.4	50.1	71.0	1860	85.3	64.9	384	85.4
Dense ViT, 10× LR	74.9	77.0	54.0	47.9	73.9	1975	86.3	70.5	481	88.6
Low-res (1/4 patches)	70.2	75.1	53.5	46.9	70.4	1942	85.3	70.5	504	86.9
Glance 25%, auxiliary loss	73.3	77.4	53.6	48.1	72.6	1883	84.6	70.3	475	87.5
Glance 25% + window	70.2	72.5	53.7	50.4	73.2	1955	85.7	70.3	501	87.8
Window-4	75.9	68.3	53.6	49.4	71.1	1934	86.0	69.9	480	87.1
Glance 25%, no auxiliary loss (ours)	75.4	77.4	53.4	50.9	73.4	1971	86.6	70.3	487	89.3
ETD, strict 5%										
Dense ViT	78.5	77.5	53.2	49.3	72.4	1842	87.6	74.3	510	89.6
Dense ViT, 10× LR	79.6	78.1	55.1	47.9	74.7	1991	88.0	75.9	585	92.2
Low-res (1/4 patches)	67.5	72.6	54.6	48.0	72.5	1977	87.6	75.4	594	89.3
Glance 25%, auxiliary loss	81.2	78.9	54.8	48.0	73.0	1943	87.6	76.1	580	91.9
Glance 25% + window	80.1	70.0	55.3	48.9	73.3	1980	87.0	75.0	562	90.5
Window-4	76.4	68.6	55.1	47.8	73.2	1960	87.7	75.0	574	89.7
Glance 25%, no auxiliary loss (ours)	82.7	78.9	53.8	49.6	75.5	1955	88.2	76.8	583	92.9
ETD, $\max(32, \lceil 0.05N \rceil)$										
Dense ViT	78.5	77.5	57.3	52.1	81.3	2092	89.2	74.3	682	95.9
Dense ViT, 10× LR	80.1	78.0	58.1	49.9	80.9	2152	88.8	76.0	709	96.6
Low-res (1/4 patches)	68.1	73.4	58.0	49.6	79.6	2162	89.1	75.6	732	94.4
Glance 25%, auxiliary loss	79.6	78.9	58.2	50.3	80.6	2138	89.1	76.1	730	97.0
Glance 25% + window	79.6	70.2	58.5	51.2	81.1	2140	89.2	74.9	688	95.4
Window-4	76.4	68.8	58.1	51.1	80.2	2095	89.0	75.1	709	94.6
Glance 25%, no auxiliary loss (ours)	82.7	78.6	57.7	51.7	81.6	2168	89.1	76.8	727	97.9

val (4,123 questions, all TextOCR word boxes) and the validation split of the scorer data (8,685 Visual-CoT and TextOCR questions, 13 sources). Per token, $\sigma(s_i)$ against "the cell touches an OCR word" (3.5M tokens, base rate 0.18) has an expected calibration error of 0.013 and a Brier score of 0.048; in the 0.3–0.8 bins it is slightly under-confident (*e.g.* 0.45 predicted, 0.53 observed). Per question, t estimates the share of the evidence map that lies on OCR cells, and on the 4,740 questions with word boxes it matches the measured share with a mean absolute error of 0.05 (Pearson 0.91). As a classifier of text-evidence questions t reaches an AUROC of 0.995 on the validation sources (median 0.87–0.96 on the text sources, 0.08–0.13 on GQA, Flickr30k, OpenImages, Visual7W and VSR), with question-level ECE 0.11: it is bimodal, and the few questions with t between 0.4 and 0.6 are text questions 90% of the time on the validation sources. t does not depend on how much text the image contains (median 0.85–0.89 from 1–5 to over 50 OCR words on TextVQA). The scorer is therefore a well-calibrated text detector; the ablations in Sec. 4.4 show that this calibration is not what drives the retention gain.

D.2. Settings of the selection-rule ablations

Every row of Tab. 3 evaluates the seed-0 model of each backbone with one switch changed at inference. The MLLM weights, the scorer, the glance threshold and K of every image are those of the full pipeline, and the Qwen and Gemma inference code implement the same switches. Every max–min rule starts from the token least similar to the mean token, as DivPrune does in the LT-OPD code.

Three training seeds. On Qwen we repeated seven rows of Tab. 3 on the seed-1 and seed-2 models. Retention over the nine benchmarks (mean ± std, change from the full pipeline): full pipeline 93.3±0.4; diversity cells 92.0±0.3 (−1.3); random cells 91.1±0.2 (−2.2); DivPrune 90.3±0.4 (−3.0); $\gamma = 1$ 92.4±0.4 (−0.9); hard split 91.7±0.5 (−1.6); DPP-r 92.5±0.8 (−0.8). The order matches seed 0, and the full pipeline is the best of the seven on every seed.

Stage 1: which cells to re-encode. For images above the glance threshold the encoder re-encodes 25% of the coarse cells at native resolution; the full pipeline takes the cells with the highest relevance r, with Stage 2 unchanged. *Max–min diversity* applies Eq. (1) to the coarse tokens, which is the rule used during training, so it measures what reading the question at inference adds. *Random cells* draws the

Table S6. Selection rule ablation on the LT-OPD + DivPrune model, identical K per image. Top: strict 5%. Bottom: $\max(32, \lceil 0.05N \rceil)$.

Method	V*	HR-4K	GQA	MMMU	MMB	MME	POPE	TextVQA	OCRB	Ret. (%)
Strict 5%										
DivPrune ($\gamma = 0$)	73.8	76.2	52.4	50.1	71.0	1860	85.3	64.9	384	85.4
Split: top-$K/2$ by p + DivPrune	78.0	75.9	52.4	50.0	70.9	1844	85.8	71.6	455	87.7
Split, scorer fine-tuned with GRPO	75.9	77.6	52.5	50.3	72.4	1887	86.1	71.7	481	88.6
Tilt set by the text confidence, $d_i(Np_i)^t$	78.0	77.6	53.2	48.7	71.8	1842	87.7	73.3	486	88.8
ETD, $d_i(Np_i)^{0.25}$	78.5	77.5	53.2	49.3	72.4	1842	87.6	74.3	510	**89.6**
$\max(32, \lceil 0.05N \rceil)$										
DivPrune ($\gamma = 0$)	73.8	76.9	57.5	51.1	80.8	2024	89.5	65.4	589	92.2
Split: top-$K/2$ by p + DivPrune	78.0	75.9	57.1	51.3	80.4	2034	88.8	71.8	652	94.2
Gate: split only if $t > 0.5$	74.3	76.4	57.4	51.7	80.9	2064	89.6	71.5	653	94.2
Soft tilt without t: $d_i(Np_i)$	78.0	77.4	55.5	51.0	79.6	2072	88.8	73.3	662	94.4
Teacher-calibrated temperature head	78.5	77.1	57.2	51.9	81.6	2077	89.2	74.0	680	95.7
Tilt $d_i(Np_i)^{\beta t}$, $\beta = 0.5$	77.0	77.2	57.5	51.8	80.5	2087	88.8	73.7	677	95.3
Tilt $d_i(Np_i)^{\beta t}$, $\beta = 1$	78.5	78.1	57.7	51.8	80.8	2090	89.2	73.2	666	95.5
Tilt $d_i(Np_i)^{\beta t}$, $\beta = 2$	79.6	78.0	57.3	50.6	80.2	2082	89.3	72.8	648	94.9
ETD, $d_i(Np_i)^{0.25}$	78.5	77.5	57.3	52.1	81.3	2092	89.2	74.3	682	**95.9**

Floor-32 rows other than $\beta = 1$ and ETD come from a separate run of the same protocol.

same number of cells uniformly, with a seed fixed per question; it keeps the zoom spread over the image but ignores the question, and tests whether coverage alone explains the gain. Both rows leave Stage 2 at ETD.

Stage 2: the two factors of ETD. All Stage-2 rows keep the top-r Stage 1. *W/o evidence* sets $\gamma = 0$, which is DivPrune on the merged native grid. *Strong evidence bias* sets $\gamma = 64$; it keeps d_i but lets p dominate. The weights $(Np_i)^\gamma$ are computed in log space, divided by their maximum and floored at e^{-80} so that they stay positive. At $\gamma = 0.25$ the floor would need Np_i to be e^{320} times below the peak and is never reached; at $\gamma = 64$ it is reached by every token whose Np_i is more than $e^{80/64} \approx 3.5$ times below the peak. The row therefore takes the tokens near the peak of p in order of p, and the remaining slots, which share the floor weight, are filled by distance. It removes diversity among the high-evidence tokens, which is where the full rule spreads the budget over a text region.

Untilted combinations. *Hard split* keeps the $\lfloor K/2 \rfloor$ tokens with the highest r and fills the other half by max–min diversity started from them, so the fill avoids tokens close to those already kept. It is the simplest way to reserve part of the budget for evidence. *Hard gate* uses the split when the text confidence exceeds 0.5 and plain DivPrune otherwise. Because t is bimodal (Sec. D.1), 0.5 separates text questions from the rest, so the row asks whether routing by question type helps; ETD has no threshold.

Tilt strength. The constant exponents 0.125, 0.5 and 1 measure the sensitivity around 0.25. The 0.5 row is not shown in Tab. 3: it gives 92.6% on Qwen and 91.7% on Gemma (-0.3 and -1.2). On the glance model (seed 0), $\gamma = 0.125$, 0.25, 0.5 and 1 retain 76.0, 77.7, 77.2 and 77.3% at a 1% budget, 92.4, 92.9, 92.6 and 92.0% at 5%, and 99.9, 99.9, 100.0 and 99.0% at 25%, so the fixed 0.25 is best at 1% and 5% and within 0.1 of the best at 25%. Since 0.25 was chosen by teacher KL on training questions, we also fixed a rule before evaluating these rows: take the γ with the lowest mean teacher KL (Eq. (3)) on 2,000 validation questions of the scorer data, with the Visual-CoT sources drawn from GQA, DocVQA and InfographicsVQA left out, using the scorer trained without the text loss. Over $\gamma \in \{0, 0.125, 0.25, 0.5, 1, 2\}$ the KL is 1.575, 1.149, 1.035, 1.031, 1.077 and 1.187, so the rule picks 0.5, and 0.5 with that scorer matches the full method over three seeds (Sec. D.3). $\gamma = t$ uses the text confidence of each question as its exponent. It is motivated by the oracle exponent on training questions, which grows with t (Sec. D.1). The AUROC of 0.90 in Sec. 4.4 treats TextVQA and OCRBench questions as reading and V*, HR-Bench 4K, GQA and POPE questions as perception (29,469 questions, glance model).

Source of p. *Wrong question* gives the Stage-2 scorer the question embedding of the preceding distinct question processed by the same evaluation worker, a question of the same benchmark; Stage 1 and the MLLM keep the true question. The row tests whether p depends on the question and not on what is salient in the image. *Question similarity* replaces the scorer by the z-scored cosine similarity between each visual token and the mean question embedding in the LLM embedding space, the relevance that CDPruner uses, and keeps the ETD rule with exponent 0.25. It is not a row of Tab. 3: it retains 73.9% with the MLLM frozen (DivPrune 75.8%, ETD 83.8%) and 87.2% on the

glance model with diversity cells (DivPrune 87.5%). *DPP-r* keeps p and replaces the greedy rule by CDPruner's conditional DPP, a greedy MAP of $\mathrm{diag}(q)\,L\,\mathrm{diag}(q)$ with L the cosine kernel of the visual tokens and quality q the min–max normalised r. It separates the scorer from the rule. With the quality $q = \exp(\alpha(r - \max r))$ instead, DPP-r retains 93.0, 92.2, 91.0, 90.5, 89.8 and 88.8% for $\alpha = 0.125, 0.25, 0.5, 1, 2, 4$ (seed 0): the flattest quality ties ETD (92.9%), so ETD matches a temperature-tuned DPP with a fixed exponent and a simpler greedy rule. Ranking the tokens by p alone (top-K) retains 88.4%, below the strong evidence bias $\gamma = 64$ (89.3%).

Rows of Tab. S6. These rows use the LT-OPD + DivPrune model (dense encoder, LT-OPD recipe) and use the exponent βt instead of a constant. *Tilt set by the text confidence* and *Tilt* $d_i(Np_i)^{\beta t}$ use the exponent βt. *Teacher-calibrated temperature head* predicts the exponent per image from t, the concentration $1 - H(p)/\log N$, the mass of the top $\max(32, \lceil 0.05N \rceil)$ tokens of p, the fraction of tokens with positive text logit and $\log N$. It chooses from $\{0, 0.25, 0.5, 1, 2, 4, 8\}$ and is trained to minimise the expected teacher KL on training questions. *Split, scorer fine-tuned with GRPO* is described in Sec. C.1.

D.3. Training-image overlap

The scorer's training images (TextVQA train and Visual-CoT, 167K images) were matched against the images of 17 evaluation sets with 64-bit difference and perceptual hashes (a pair matches when the dHash distance is at most 6 and the pHash distance at most 10). Evaluation images with at least one match (questions on them in parentheses): POPE 132 / 500 (2,376 / 9,000; COCO images), DocVQA 105 / 1,284 (491 / 5,349), MMBench 47 / 1,115 (187 / 4,329), TextVQA 70 / 3,166, MME 24 / 1,177 (50 / 2,374), OCRBench 34 / 930 (35 / 1,000) and GQA 1 / 398. The deduplicated scorer is trained identically on data from which every training image matching an evaluation image is removed (851 images, 1,720 of 422K records, mostly Visual-CoT GQA, Visual7W and DocVQA; the removal rule also adds a pixel-correlation and a 256-bit dHash test). With the deduplicated scorer in both stages, ETD at strict 5% retains, on the nine LT-OPD benchmarks, 84.2% on the untrained MLLM (83.8% with the original scorer) and 92.0±0.1% on the earlier glance model over three seeds (92.0±0.1%). DPP-r with the deduplicated scorer retains 83.1% on the untrained MLLM (82.5%), so ETD stays 1.1 points ahead; with the original scorer the paired difference is +1.3 (95% interval $[0.4, 2.2]$), mostly from GQA. On the five additional benchmarks of Tab. 1, HR-Bench 8K, MME-RealWorld, HRScene, DocVQA and InfoVQA, it retains 78.5% (78.9%) on the untrained MLLM, 87.7±0.5% (87.5±0.6%) on the glance model over three seeds, 84.2% (83.4%) on the LT-OPD-recipe model and 87.1% (86.6%) on the 10× dense-encoder model; on the trained models no benchmark moves by more than 0.8. The LT-OPD-recipe model with ETD scores 76.5 (75.9) on DocVQA, so its gain over DivPrune (52.5) is 24.0, not 23.4. The overlap does not inflate the reported results on these sets. A stricter scorer is trained without every source that shares a dataset with an evaluation benchmark (the TextVQA, TextCaps, DocVQA, InfographicsVQA and GQA parts removed; 218K records kept). On the nine benchmarks it retains 84.0% (83.8%) on the untrained MLLM and 93.0±0.5% (93.3±0.4%) with Glance-ETD over three seeds. On the five additional benchmarks it retains 78.2% (78.9%) and 87.3±0.6% (88.5±0.4%); the loss is on DocVQA (82.6→80.1) and InfoVQA (48.1→45.4), whose document training data was removed. A scorer retrained on the features of each trained model, with the exponent βt instead of the constant 0.25, gave 91.5% on the dense-encoder model (91.9% with the shared scorer, seed 0) and $90.0 \pm 0.4\%$ on the glance model ($90.3 \pm 0.8\%$), so one shared scorer is used for all Qwen models. The scorer also needs little box supervision: trained on random 5%, 15% or 40% of the 422K records with everything else fixed, ETD on the untrained MLLM retains 82.7, 84.2 and 83.9% (83.8% with all records; DocVQA 65.7, 70.7 and 72.4 against 71.9), so about 63K records suffice and 5% loses about one point, mostly on reading.

Same supervision for every selector. Tab. S7 trains the same 1.9M scorer under the supervision of the learned baselines and runs top-K, DPP-r and ETD on each, with the MLLM frozen. With 14K box records, as many as LT-OPD-14K, ETD retains 81.1%, two thirds of the gain over DivPrune of the full scorer; without the question input, which makes it a box-supervised LearnPruner-style scorer, 81.3%, so 5.5 of the 8.0 points come from a question-independent evidence prior (text and salient objects) and 2.5 from reading the question. On Glance-ETD the question-independent scorer, used in both stages, retains 92.3% over three seeds, against 93.3% with the full scorer and 89.4% with DivPrune, so it keeps 2.9 of the 3.9 points (86.2 *vs.* 88.5% on the five additional benchmarks). Trained on LT-OPD-14K with the LearnPruner objective and no boxes, the scorer gives no gain under any selector (ETD 75.8%, equal to DivPrune). The gain therefore comes from box supervision, not from the size of the scorer or the amount of data. With a box-supervised scorer, top-K stays 5–8 points below DPP-r and ETD, and ETD is ahead of DPP-r except with 14K records, where the two are within 0.4.

Neither the text head nor in-domain reading supervision is needed. On the glance model with diversity cells (three seeds), a scorer trained without the text loss, used with $\gamma = 0.5$ as chosen in advance by the validation-KL rule (Sec. 4.4), retains 90.8%, the same as the full scorer at $\gamma = 0.25$ (90.8%; paired difference -0.05, 95% inter-

Table S7. Scorer supervision and selector with the MLLM frozen, strict 5%, retention over the nine LT-OPD benchmarks (DivPrune: 75.8%). All scorers have the same 1.9M-parameter architecture; 14K = as many box records as LT-OPD-14K; LearnPruner = its objective on LT-OPD-14K.

Scorer	Data	Boxes	Question	Top-K	DPP-r	ETD
Ours	422K	✓	✓	77.9	82.5	83.8
14K box records	14K	✓	✓	73.6	81.5	81.1
No question	422K	✓	–	73.9	80.1	81.3
No boxes (LearnPruner)	14K	–	✓	57.8	74.1	75.8

val $[-0.57, 0.45]$) and 3.1 points above DivPrune. Also removing TextVQA and TextCaps [45] from its training data changes retention by $+0.3$ on that model and $+0.6$ on the untrained MLLM (seed 0, both intervals contain zero).

D.4. Stage-1 cell choice

Tab. S8 changes only how the glance encoder picks its 25% of coarse cells, at inference, on all three training seeds; the trained weights, zoom ratio, second pass and token budget are unchanged, so the encoder FLOPs are the same in every row, and Stage 2 is ETD in every row. *Top-p* (ours) takes the cells with the highest scorer relevance; *diversity* is max–min selection on the coarse tokens, the rule used in training; *query-aware* runs ETD on the coarse tokens; *half top-p* takes half of the cells by relevance and the rest by diversity. R_{fine} is the share of answer-box cells re-encoded at native resolution on the 345 TextVQA questions whose image takes the glance path. Over the nine benchmarks 6.1% of the images (60% of the visual tokens) take the glance path: all of V* and HR-Bench 4K, 25% of MME, 15% of OCRBench, 10% of MMMU, 8% of TextVQA, and none of GQA, MMBench or POPE, so the nine-benchmark gain of 1.3 points over diversity comes from few benchmarks (V* 77.7→83.6, OCRBench 576→590, three-seed means).

Diversity covers text evidence about as well as random cells (32% *vs*. 26%), and on documents it is below random. Every rule that reads the scorer raises the mean of the four additional benchmarks other than HR-Bench 8K (H4) from 56.5 to 58.2–59.0, mostly on documents and on MME-RealWorld. The seed-to-seed spread of this mean is at most 0.7. Random cells raise DocVQA too but are 8 points below top-p on HR-Bench 8K, so coverage alone does not explain the gain. On the earlier glance encoder (Sec. C.2), taking the cells that touch the answer box first (an oracle, seed 0, with the βt exponent in Stage 2) re-encodes 99% of them and gains a further 0.4 on glanced TextVQA questions over query-aware cells. The scorer adds about 2 ms per image, and the time to first token on HR-Bench 4K is 1.28 s against 1.39 s for diversity (full tokens: 4.15 s). With scorer-chosen cells the glance model's DocVQA (82.6) is above the dense-encoder model with ETD (80.1, three seeds), so the document gap of a glance encoder with diversity cells comes from the cell choice, not from encoding a quarter of the image.

Zoom ratio and threshold. This paragraph and the next use the earlier glance encoder. On seed 0, re-encoding 12.5%, 25% or 50% of the cells gives an H4 mean of 55.3 / 55.0 / 57.8 with diversity and 56.0 / 58.2 / 58.9 with query-aware cells, with a time to first token on HR-Bench 4K of about 1.2–1.3 s at 12.5% and 2.0 s at 50%. Scorer-chosen cells at 25% thus match diversity at 50% for less than three quarters of its encoder time. Applying the glance to every image instead of those above 4096 patches lowers the nine-benchmark retention of the query-aware rule from 91.4% to 89.6% (TextVQA 71.3), so small images stay on the dense path instead of the glance. The threshold is not sensitive: moving it at inference to 2048, 3072 or 6144 patches (trained at 4096) gives Glance-ETD 91.3, 91.4 and 91.8% on the nine benchmarks (91.9% at 4096, seed 0) and 87.9, 87.8 and 88.3% on the five additional benchmarks of Tab. 1 (88.0%); lower thresholds send mid-sized reading images to the glance path (TextVQA 76.1→74.0 at 2048).

Coarse and merged features. The scorer is trained on dense features but reads the coarse grid in Stage 1 and the merged grid in Stage 2. A Stage-1 scorer trained on 2×2-pooled features gives 92.0, 91.9 and 92.4% over the three seeds, against 91.9, 92.0 and 92.2% with the shared scorer. Filling the unselected cells by bilinear upsampling of the coarse tokens instead of copying each to its four positions gives 91.5% at inference and 91.6% when also used in training (copying: 91.9%, seed 0). The feature shift does not cost accuracy, and copying is kept.

E. Budgets and Diagnostics

The token budget is varied around the LT-OPD operating point, and the evidence checks include Gemma 4.

E.1. The LT-OPD operating point

The LT-OPD release sets $K = \max(32, \lceil 0.05N \rceil)$. Small images hit the floor of 32 tokens, so the nine benchmarks together keep 7.1% of the visual tokens, not 5.0%. Tab. S9 evaluates each method at this budget. LT-OPD retains 86.1%; training with DivPrune on the same data reaches 92.2%. Glance-ETD retains $98.0 \pm 0.1\%$ over three seeds, 11.9 points above LT-OPD with the same number of tokens, and the dense-encoder variant reaches $96.8 \pm 0.2\%$. The released LT-OPD checkpoint, which keeps the learnable token merge, retains 91.1% (Sec. B.4).

E.2. Sweep across Token Budgets

Tab. S10 varies the strict budget from 1% to 25%. On the untrained model ETD is ahead of DivPrune at every measured budget (+10.3 at 1%, +8.0 at 5%, +6.4 at 10%, +3.7

Table S8. Stage-1 cell rules at inference on the glance model at strict 5% (mean of three training seeds; Stage 2 is ETD in every row). R_{fine}: share of answer-box cells of the 345 glanced TextVQA questions that are re-encoded at native resolution (‡earlier glance encoder, seed 0). HR-8K of random: seed 2 only. Ret.: nine LT-OPD benchmarks. H4: mean of DocVQA, InfoVQA, MME-RealWorld and HRScene. TTFT: HR-Bench 4K, batch 1, one A800 (full tokens 4.15 s); timed with the earlier weights, which do not change the compute; half top-p was timed in a second run and scaled by its top-p time.

Stage-1 cells	R_{fine}	Ret.	HR-8K	DocVQA	InfoVQA	MME-RW	HRScene	H4	TTFT (s)
Top-p (ours)	88‡	**93.3**	**75.9**	**82.6**	**48.1**	**48.8**	56.7	**59.0**	1.28
Diversity (as trained)	32	92.0	70.3	79.5	44.8	46.0	55.6	56.5	1.39
Random	26	91.1	67.9	80.2	46.3	44.8	55.4	56.7	1.31
Query-aware	73	93.0	74.9	81.4	46.6	48.0	**56.8**	58.2	1.42
Half top-p	79	92.9	74.4	81.8	47.1	48.0	56.4	58.3	1.45

Table S9. Operating point of the LT-OPD release, $K = \max(32, \lceil 0.05N \rceil)$ per image ($\sum K / \sum N = 7.1\%$). Rows above the line are baselines and LT-OPD-recipe models with DivPrune; rows below are ours.

Method	V*	HR-4K	GQA	MMMU	MMB	MME	POPE	TextVQA	OCRB	Ret. (%)
Full tokens	80.6	78.0	60.0	51.3	79.0	2218	89.9	80.5	848	100.0
DivPrune [2], training-free	70.7	73.9	53.4	48.8	75.5	1925	86.6	49.3	431	84.1
CDPruner [64], training-free	53.4	62.4	40.8	48.2	74.2	1862	81.9	43.5	398	75.4
LT-OPD [26] (arXiv'26.09)	67.0	69.6	51.7	49.4	77.8	2073	85.9	56.1	533	86.1
LT-OPD + DivPrune (ours, reproduced)	73.8	76.9	57.5	51.1	80.8	2024	89.5	65.4	589	92.2
Ours and its dense-encoder variants										
Glance-ETD (ours)	83.8	78.4	58.1	51.7	81.4	2152	89.1	76.4	727	**98.0**±0.1
ours, dense-encoder variant (no glance)	79.1	77.1	58.1	50.4	80.7	2160	89.1	76.4	723	96.8±0.2

The full-token row is the non-thinking run that every retention in the paper is normalised by.

Table S10. Retention (%) versus strict budget $\sum K / \sum N$.

	1%	2%	3.5%	5%	7.5%	10%	15%	25%
Untrained								
ETD	66.3	75.2	81.0	83.8	87.3	90.1	93.0	95.7
DivPrune	56.0	64.6	71.5	75.8	80.9	83.7	87.6	92.0
DART	49.2	55.5	62.4	67.7	73.2	77.2	84.4	91.0
CDPruner	51.3	58.3	63.7	67.6	71.5	74.3	78.6	84.4
PruneSID	51.7	59.2	65.1	68.7	77.0	80.5	81.7	86.0
ZOO-Prune	52.9	61.2	67.2	70.9	76.2	79.3	83.4	90.3
LT-OPD								
ETD	74.2	81.6	86.7	89.6	93.1	95.1	97.1	99.4
Glance								
ETD	77.7	85.3	90.0	92.9	95.6	97.1	98.4	99.9
DivPrune	68.5	77.9	85.4	89.3	92.8	94.3	96.5	98.2

at 25%) and ahead of DART by a wider margin. CDPruner, PruneSID and ZOO-Prune also stay below DivPrune at 10% (74.3–80.5) and 25% (84.4–90.3). ETD at 10% (90.1%) is within 1.9 points of DivPrune at 25% (92.0%) with 2.5× fewer tokens. After training, ETD at 3.5% (86.7%) is above DivPrune at 5% (85.4%). On the glance model, where the ETD row also uses scorer-chosen cells and the DivPrune row diversity cells, the gap falls from 9.2 points at 1% to 3.6 at 5% and 1.7 at 25%: once a quarter of the tokens is kept, diversity alone already covers most evidence.

Table S11. Global budget allocation on the LT-OPD model with ETD. The power of N and the open-prompt factor are fitted on training questions only; the scale is set to match the reference $\sum K / \sum N$ over the benchmarks shown below.

Allocation	strict 5%	floor-32 budget (7.1%)
Same ratio for every image	89.6	95.9
$K \propto N^{0.45}$	94.6	96.3
$K \propto N^{0.45} \cdot 2^{[\text{open}]}$	**95.1**	**97.3**

E.3. Allocating the budget across images

Tab. S11 applies Eq. (S1) on top of ETD on the dense LT-OPD model. With the same total number of tokens ($\sum K / \sum N$ of 5.00% and 4.98%), giving large images a smaller fraction and small images a larger one raises strict-5% retention by 5.0 points, and by 5.6 with the open-prompt factor. The gain comes from the small-image benchmarks (OCRBench 510→732, MMBench 72.4→79.0, MME 1842→2094), while V*, whose images are the largest, loses 3.6 points. At the floor-32 budget the gain is smaller (+1.4), because the floor already gives small images more than 5%. In an offline sweep of η composed from the fixed-ratio runs, the training value 0.45 is also the best evaluation value. On Glance-ETD

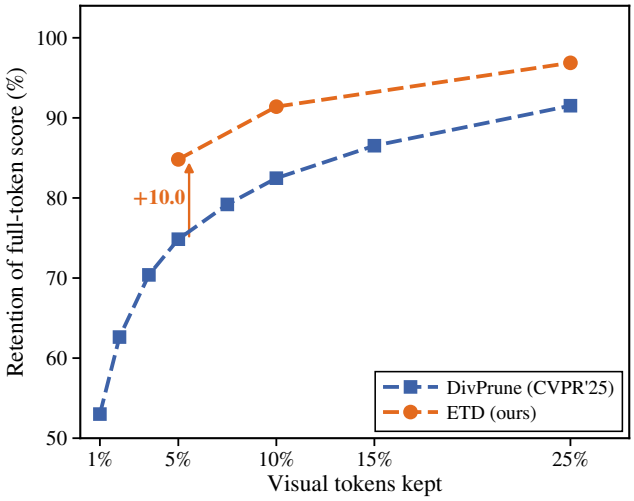


Figure S1. Retention of Gemma 4 E4B-it against its own full-token scores. ETD and DivPrune pass the same number of visual tokens per image. Every marker averages all nine benchmarks.

(seed 0) the same allocation raises strict-5% retention from 92.9 to 97.0, and to 97.7 with the open-prompt factor, mostly on OCRBench (583→718 and 781), MMBench (75.5→80.4) and GQA (53.8→57.7); on training-free ETD with Gemma 4 it is neutral (85.1 *vs*. 85.2).

E.4. Evidence diagnostics

Tab. S12 tests whether ETD's gain comes from the evidence the question asks for. Scoring tokens with a different question lowers TextVQA from 74.3 to 68.9 and OCRBench from 510 to 441 on the LT-OPD model, and from 76.8 to 72.7 and from 583 to 523 on Glance-ETD, so the scorer uses the question. It still stays above DivPrune with the wrong question (87.5 *vs*. 85.4% on the LT-OPD model, 90.7 *vs*. 90.0% on Glance-ETD with the same Stage 1), because part of the evidence map does not depend on the question (Sec. D.3). Removing the tokens ETD would have chosen and selecting from the rest is worse than DivPrune (TextVQA 61.6 *vs*. 64.9 and 69.0 *vs*. 70.3), so ETD's tokens carry information that the rest of the image does not replace. With 0.1% of the tokens the models fall to 55–57%, close to answering from the question alone with no image, so retention is informative between about 55 and 100%.

E.5. Gemma 4 across budgets

Fig. S1 extends the Gemma comparison past strict 5%. ETD retains 91.4% at 10%, against 82.5% for DivPrune, and 96.9% at 25% (DivPrune 91.5%); ETD at 10% matches DivPrune at 25%. At 5%, TextVQA rises from 38.9 to 59.9 and OCRBench from 184 to 396, against a full-token OCRBench of 764, and HR-Bench 4K rises from 53.0 to 62.0. GQA, MMMU and POPE stay within 1.0 point of DivPrune. On Gemma, Glance-ETD uses the two-pass glance with Gemma's own vision tower: a quarter-resolution pass gives coarse features for every cell, the Gemma scorer picks the 25% of the cells with the highest relevance, and only those are re-encoded at native resolution before ETD selects K tokens. It runs on the self-distilled checkpoints, which are trained without the glance, so unlike Qwen the encoder is not adapted to the glance features. Over three seeds it retains 91.1±2.0% at strict 5%, against 91.3±1.1% for ETD with the dense encoder and 87.2 ± 1.2% for DivPrune; with diversity cells instead of scorer-chosen ones it retains 89.7 ± 1.2%.

E.6. Other MLLMs

Tab. S13 repeats the strict-5% comparison on GLM-4.6V-Flash and InternVL3.5-8B, next to Qwen3.5-4B and Gemma 4 E4B-it. Each model has its own scorer, trained on the same box data from the features of that untrained model, and is self-distilled with the recipe of Sec. 3.5 for 175 steps, once with the dense encoder and once with the glance encoder. InternVL splits an image into crops: its glance encoder encodes the thumbnail, fills every crop with the upsampled thumbnail region, and re-encodes at full resolution the 25% of the crops to which the scorer gives the highest mean relevance. GLM and InternVL rows use one training seed, GQA is scored with the LLaVA answer normalisation on every row of these two models, and MMBench groups are adjudicated as in Sec. B.2.

ETD is ahead of DivPrune on every model, by 8.0–13.0 points with the MLLM untrained and by 3.6–12.0 after self-distillation; the largest gaps are on InternVL, where DivPrune keeps 150 of 800 points on OCRBench and ETD 374. Self-distillation adds 3.9–8.7 points to ETD. The glance encoder is 0.8 points above the dense one on Qwen and 0.2, 1.1 and 2.4 points below it on Gemma, GLM and InternVL. On GLM the loss is on V* (67.5 *vs*. 74.9) with HR-Bench 4K level (71.4 *vs*. 71.7); on InternVL it is on V* and HR-Bench 4K (53.4 *vs*. 64.9, 55.4 *vs*. 63.0), while POPE is higher with the glance encoder (72.9 *vs*. 67.2).

F. Latency and Memory

Accuracy against cost. Matched to the benchmark that is timed (Fig. S2a,b), Glance-ETD keeps 78.9 on HR-Bench 4K and 82.7 on V* (seed 0) at 2.9× and 1.4× TTFT, while the low-resolution and windowed encoders give speed for detail. Along the budget axis (Fig. S2c) a dense-encoder model moves by at most 30 of its 370–400 TFLOPs between 1% and 25%, because the encoder dominates the cost; the glance encoder moves the whole curve to 60–90 TFLOPs, and Glance-ETD at 7.5% (95.6%) uses a fifth of the FLOPs of the dense model at 5%.

Table S12. Does ETD keep the evidence? Strict 5%. *Wrong question*: ETD scores tokens with the question of another sample, the model then answers the true question. *Complement*: ETD-selected tokens are removed from the pool before selecting. *Blind*: 0.1% of tokens.

Method	V*	HR-4K	GQA	MMMU	MMB	MME	POPE	TextVQA	OCRB	Ret. (%)
LT-OPD + DivPrune model										
DivPrune	73.8	76.2	52.4	50.1	71.0	1860	85.3	64.9	384	85.4
ETD	78.5	77.5	53.2	49.3	72.4	1842	87.6	74.3	510	89.6
ETD, wrong question	77.5	78.0	51.3	49.7	71.4	1871	87.0	68.9	441	87.5
ETD complement	72.3	77.4	51.9	49.4	72.5	1907	85.7	61.6	393	85.3
Blind (0.1%)	51.8	59.6	35.4	46.6	46.1	1314	61.4	10.9	59	55.2
Glance model (top-p Stage 1 in the ETD rows)										
DivPrune	75.4	77.4	53.4	50.9	73.4	1971	86.6	70.3	487	89.3
ETD	82.7	78.9	53.8	49.6	75.5	1955	88.2	76.8	583	92.9
ETD, wrong question	80.6	78.8	52.5	49.1	74.5	1956	87.7	72.7	523	90.7
ETD complement	74.3	79.1	52.9	49.9	73.6	1896	86.8	69.0	490	88.6
Blind (0.1%)	49.7	62.2	35.1	45.8	51.1	1333	63.7	13.3	68	56.6

Table S13. Retention over the nine LT-OPD benchmarks at strict 5% on four MLLMs (Fig. 5 plots GLM and InternVL). Each model is compared with its own full-token scores. Qwen and Gemma trained rows are means of three seeds; GLM and InternVL use one seed.

Method	Qwen3.5-4B	Gemma 4 E4B-it	GLM-4.6V-Flash	InternVL3.5-8B
DivPrune	75.8	74.8	74.7	69.4
ETD	83.8	84.8	84.3	82.4
Self-distillation + DivPrune	88.8	87.2	83.5	74.2
Self-distillation + ETD	92.5	**91.3**	**89.3**	**86.3**
Glance-ETD	**93.3**	91.1	88.2	83.8

Table S14. Efficiency on one A800-SXM4-80GB (bf16, median over 30 questions, configurations run one after another). HR-4K unless noted; batch 1 unless noted. Times are milliseconds. TFLOPs are ViT + LLM, attention included. Speed-ups over full tokens are in parentheses. Peak memory is batch 1 / batch 8.

	Full	Dense + ETD	Glance-ETD
TTFT	3955	3305 (1.20×)	1376 (**2.87**×)
V* TTFT	547	476 (1.15×)	379 (1.44×)
End-to-end	4246	3609 (1.18×)	1683 (2.52×)
TFLOPs	366 + 116	366 + 6	59 + 6
Batch-8 q/s	0.277	0.342 (1.23×)	0.815 (2.94×)
Mem. b1/b8 (GB)	13.3 / 34.7	13.7 / 35.1	12.7 / **17.7**

Where the speed-up appears. On the twelve benchmarks the TTFT speed-up grows with the number of visual tokens, as the encoder takes a larger share of the cost: 1.5× on V* and DocVQA and 2.5–2.9× on HR-Bench (Fig. S2j). End-to-end speed-ups are smaller because decoding takes the same 59 ms per token with or without pruning at batch 1 (Fig. S2e,k, same decode stage).

FLOPs. We count all matrix multiplications of one TTFT call, including the attention products, from the checkpoint shapes and the measured token counts of each question: ViT blocks $2(4d^2 + 2dd_{\text{ff}})$ per patch plus $4N^2d$ for global attention (d=1024, d_{ff}=4096, 24 blocks), the merger, and the LLM as two FLOPs per non-embedding parameter (3.57B) per token plus causal attention in its eight full-attention layers; the Gated-DeltaNet recurrence and the selector are below 0.5%. On HR-Bench 4K global attention is 332 of the 366 ViT TFLOPs (Fig. S2d). Counting only the linear layers gives 138, 41 and 24 TFLOPs for full tokens, dense + ETD and Glance-ETD, which hides the encoder's quadratic cost. Measured speed-ups stay well below the FLOPs ratio (2.51× against 7.5× at 8K, Fig. S2f): the LLM prefill alone is 9.8× faster, but the ViT only 4.0×, and host-side preprocessing does not shrink (Fig. S2e).

Memory. Weights take 9.1 GB. At batch 1 the remaining peak is set by ViT activations, so selection after the encoder does not lower it (selection itself adds about 0.4 GB on HR-Bench 4K). Per request, the decode cache is 32 KiB of KV per kept token in the eight full-attention layers plus a constant 49 MiB Gated-DeltaNet state (24 layers, fp32 state assumed); at 5% the KV part falls from 448 to 25 MiB on HR-Bench 4K, so the hybrid architecture limits the cache saving to 6.7× (Fig. S2l). Peak memory at batch 8 grows with the image for full tokens and the dense encoder, to 35 GB at 8K, and stays below 18 GB with the glance encoder (Fig. S2h).

Tab. S14 collects the numbers behind Fig. 4 on HR-Bench 4K and V*. Tab. S16 times every Tab. 1 method on Qwen3.5-4B in a separate run with the same protocol. All of them prune after the dense encoder, so the ViT bounds their speed-up; methods with a heavy selection step (CD-Pruner, PruneSID, ZOO-Prune and LT-OPD, whose DPP takes 0.6 s at 16k tokens) or extra ViT work (VisionTrim) fall to 0.87–1.09× against full tokens.

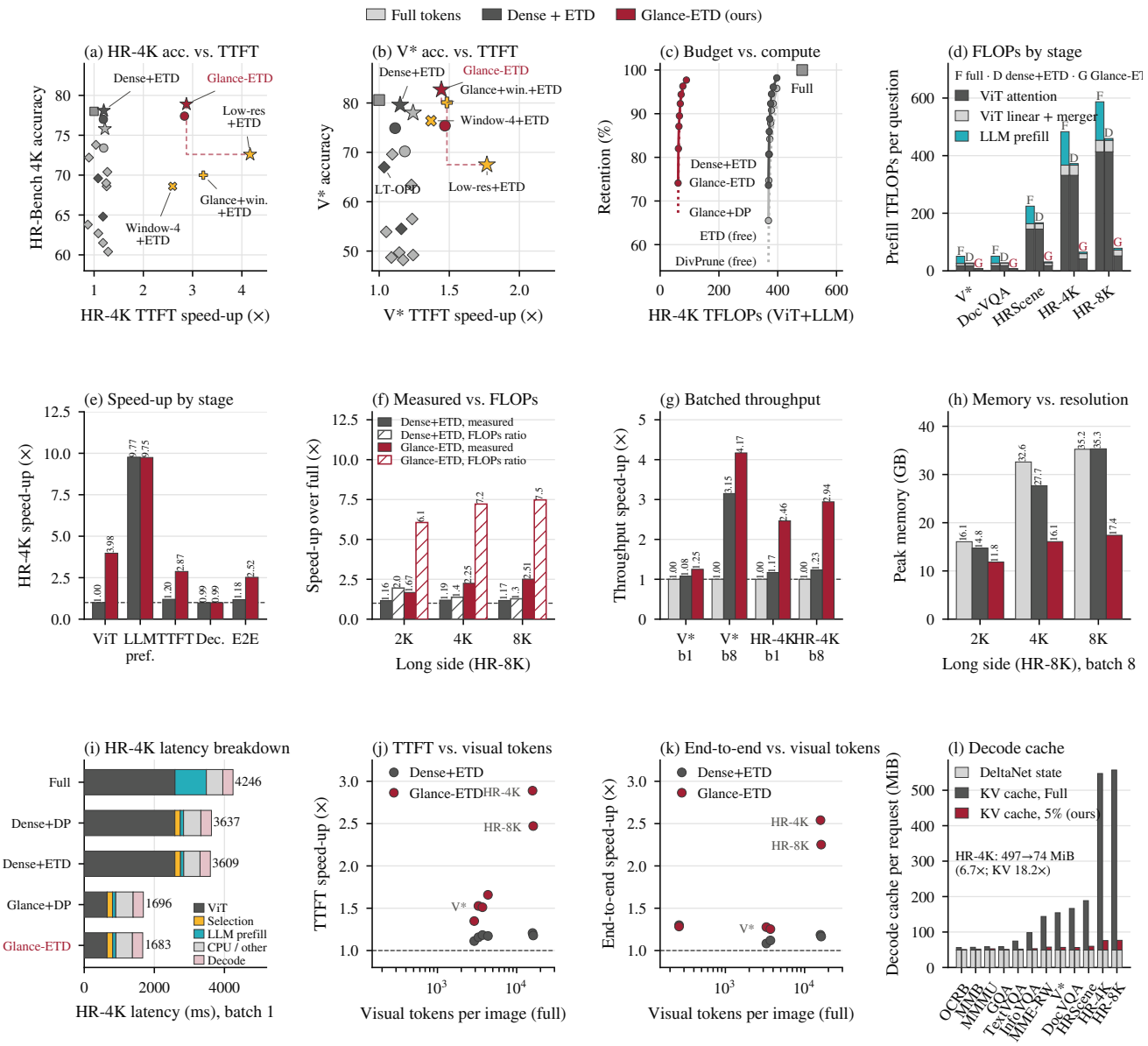


Figure S2. Further efficiency views (strict 5%, Qwen3.5-4B). (a)(b) Task-matched trade-off: HR-Bench 4K and V* accuracy against TTFT speed-up on the same benchmark (markers as in Fig. 4a). (c) Retention against HR-Bench 4K prefill TFLOPs as the budget goes from 1% to 25% (Tab. S10). (d) Prefill FLOPs per question by stage (F full, D dense + ETD, G Glance-ETD). (e) HR-Bench 4K speed-up of each stage (ViT, LLM prefill, TTFT, decoding, end to end). (f) Measured TTFT speed-up against the FLOPs ratio (hatched) on HR-Bench 8K resized to 2K–8K. (g) Throughput speed-up at batch 1 and 8 (Tab. S18). (h) Peak memory at batch 8 on resized HR-Bench 8K. (i) HR-Bench 4K latency by stage at batch 1 for all five configurations (end-to-end ms at the bar ends). (j)(k) TTFT and end-to-end speed-up against the mean number of visual tokens, on the benchmarks where Glance-ETD exceeds 1.15×. (l) Decode cache per request: KV of the eight full-attention layers plus the constant Gated-DeltaNet state. (a)–(c) and (e)–(i) use the run of Tab. S14 for full tokens and the dense and glance encoders; (d) and (j)–(l) use a 4-GPU run that covers all 12 benchmarks, with speed-ups taken against its own full-token run.

Protocol. All latency numbers come from one run per image size on a single A800-SXM4-80GB (driver 550.54.15, PyTorch 2.10 with CUDA 12.6, FlashAttention [10], bf16), with the five configurations (full tokens, dense encoder with DivPrune or ETD, glance encoder with DivPrune or ETD) run one after another on the same otherwise idle GPU. Each configuration sees the same questions (30 per benchmark at native size, 24 HR-Bench 8K questions per resized size) after three warm-up questions. TTFT is split by CUDA-synchronised hooks into ViT, scorer, selection,

Table S15. Prefill (TTFT) and end-to-end (E2E) speed-up over full tokens on every benchmark with a latency measurement, strict 5% (batch 1, median over 30 questions, one A800). Qwen3.5-4B: same runs as Tab. S14; Gemma 4 E4B: trained Glance-ETD model (Sec. E.5), about 1.1k visual tokens per image, configurations interleaved per question. Fig. 4c,d shows, per panel, the three benchmarks with the largest Glance-ETD speed-up of each model.

	Qwen3.5-4B				Gemma 4 E4B			
	Dense + ETD		Glance-ETD		ETD		Glance-ETD	
Benchmark	TTFT	E2E	TTFT	E2E	TTFT	E2E	TTFT	E2E
HR-4K	1.20	1.18	2.87	2.52	1.05	1.04	1.13	1.11
HR-8K	1.18	1.17	2.42	2.23	1.26	1.03	1.15	0.93
V*	1.15	1.09	1.44	1.25	1.07	1.03	1.33	1.23
DocVQA	1.19	1.12	1.38	1.20	1.05	1.24	1.37	1.36
HRScene	1.20	1.14	1.37	0.96	1.05	0.82	1.35	0.96
MME-RW	1.10	1.04	1.22	1.16	1.24	1.07	1.51	1.23
GQA	0.93	1.11	0.89	1.28	–	–	–	–
InfoVQA	0.93	1.08	0.89	1.02	1.04	1.06	1.39	1.25
MMBench	0.92	0.98	0.88	0.96	–	–	–	–
TextVQA	0.92	0.98	0.87	0.97	1.04	1.04	1.33	1.24
MMMU	0.92	0.99	0.87	0.95	–	–	–	–
OCRBench	0.91	1.27	0.84	0.99	1.06	1.41	1.40	1.12

Table S16. Latency of the Tab. 1 methods on Qwen3.5-4B at strict 5% (one A800, bf16, batch 1, median over 30 questions, configurations run one after another; speed-ups over full tokens of the same run). Gain is the speed-up of TTFT over full tokens. TTFT and Sel. (selection) in ms, Mem. is HR-Bench 4K peak memory in GB. Sel. is the selection time on HR-Bench 4K (our implementations; DivPrune and ETD use a CUDA-graph greedy). VisionTrim recomputes a ViT attention map, which the fused attention kernel does not return (1.5 s of extra ViT time); ZOO-Prune spends 1.35 s on 64 zeroth-order probes of the projector. Ret. is Ret. All of Tab. 1 (fourteen benchmarks). The Glance-ETD row was timed with diversity-chosen glance cells; choosing them with the scorer adds one scorer call of about 2 ms (Sec. F).

		HR-Bench 4K			V*	
Method	Ret.	TTFT	Gain	Sel.	Gain	Mem.
Full tokens	100.0	4099	–	0	–	13.0
DivPrune	71.3	3430	1.19×	141	1.18×	13.2
DART	62.8	3469	1.18×	266	1.14×	13.1
CDPruner	61.8	3776	1.09×	592	1.05×	13.0
PruneSID	65.2	3948	1.04×	769	1.09×	13.3
VisionTrim	56.9	4696	0.87×	12	1.09×	14.7
HAWK	52.9	3190	1.28×	14	1.24×	13.2
ZOO-Prune	67.0	4576	0.90×	1352	0.76×	13.2
LLaVA-Mini	66.8	3267	1.25×	12	1.24×	13.1
LearnPruner	58.8	3290	1.25×	39	1.17×	13.1
MetaCompress	65.9	3228	1.27×	15	1.23×	13.1
EPIC	63.0	3471	1.18×	263	1.16×	13.1
LT-OPD	74.3	3791	1.08×	581	1.04×	13.1
ETD (ours)	82.0	3362	1.22×	148	1.24×	13.2
Glance-ETD (ours)	91.6	1413	2.90×	162	1.48×	12.2

LLM prefill and the remainder (CPU preprocessing and other host work); FLOPs are counted analytically as described below. A second run of the 4K setting on another GPU reproduces every median within 2%. Full-token p95 on HR-Bench 4K is inflated by two questions whose prefill takes about 15 s, which we attribute to kernel compilation for unseen sequence lengths; the medians are unaffected. The timed glance encoder picks its cells by diversity. A separate run times the final model with scorer-chosen cells, the configuration of the paper, together with full tokens, diversity cells and the dense encoder with ETD (six benchmarks, 30 questions each, one A800). The scorer call takes about 2 ms. HR-Bench 4K TTFT is 1283 ms against 3964 ms for full tokens (3.09×), 1367 ms with diversity cells (2.90×) and 3318 ms for the dense encoder with ETD; V* is 1.47× faster, and TextVQA, GQA, MMBench and OCRBench run at 0.88–0.90×. The speed-ups reported here are therefore not overstated.

Tab. S17 varies the image size. The speed-up grows with resolution, from 1.67× at 2K to 2.51× at 8K, as the encoder saving dominates; at 8K CPU preprocessing (865 ms) becomes the largest remaining cost. The scorer itself takes 2–3 ms at every size; most of the selection time is the greedy loop. Tab. S18 shows that batching does not help the full-token model at these sizes (its throughput stays flat or drops), while the pruned models gain from batching and Glance-ETD halves the batch-8 memory on HR-Bench 4K.

Table S17. Image size. HR-Bench 8K images resized to a long side of 2K–8K pixels (24 questions, batch 1). Glance-ETD TTFT is split into ViT, scorer plus selection (Sel.), LLM prefill and CPU; speed-up is full / Glance-ETD at the median. Memory is peak allocated GPU memory of that configuration on the single A800 used for the timing run. Times are in ms.

Long side	2K	4K	8K
Visual tokens N	2,688	10,752	16,224
Full TTFT p50/p95	385 / 532	2283 / –[‡]	4511 / 4719
Dense + ETD TTFT	331	1913	3845
Glance-ETD p50/p95	230 / 384	1015 / 1801	1797 / 1855
ViT/Sel./LLM/CPU	71/20/60/78	402/127/91/392	678/161/91/865
Speed-up	1.67×	2.25×	**2.51**×
Mem. full/glance (GB)	10.8 / 10.9	12.4 / 12.2	13.4 / 12.3

[‡] 15.9 s, from the compilation outliers described in the text.

Table S18. Throughput (questions/s, generation up to the official answer cap or 128 tokens) at batch 1, 4 and 8, and peak memory at batch 8; same run as Tab. S14. Speed-ups relative to full tokens in parentheses next to each throughput.

	Batch	Full	Dense + ETD	Glance-ETD
HR-4K	1	0.265	0.310 (1.17×)	0.653 (2.46×)
	4	0.280	0.343 (1.23×)	0.799 (2.85×)
	8	0.277	0.342 (1.23×)	0.815 (**2.94**×)
	mem. (GB)	34.7	35.1	17.7
V*	1	1.169	1.264 (1.08×)	1.462 (1.25×)
	4	0.910	2.229 (2.45×)	2.873 (3.16×)
	8	0.826	2.602 (3.15×)	3.444 (**4.17**×)
	mem. (GB)	17.5	16.0	12.5